# The CMB temperature angular power spectrum through fourth order in a radiation-only universe: exact observable, initial data, perturbative kernels and harmonic machinery

(collaborative derivation; see the verification ledger, App. B)

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**Abstract**

We set up the calculation of the observed temperature angular power spectrum $C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}] + O(\epsilon^6)$ in a spatially flat, radiation-only universe governed by general relativity with $\Lambda = 0$, for photons emitted on the physical hypersurface $T_{\rm phys} = T_*$ and observed by a tagged fluid element at fixed proper time. The settled parts are presented here: (i) an exact, non-perturbative reformulation of the measurement, $T_{\rm obs}(\hat{\mathbf{n}}) = T_o\, e^{-S(\hat{\mathbf{n}})}$ with $S$ the fluid shear integrated along the ray, which contains all Sachs–Wolfe, Doppler, integrated Sachs–Wolfe, lensing, time-delay, emission-surface and observer physics; (ii) the definition of the Gaussian non-linear primordial variable and the construction of regular initial data satisfying the constraints at every order; (iii) the perturbative solution of the Einstein–fluid system through third order in Poisson gauge (closed first-order transfer functions, second-order scalar/vector/tensor kernels and Green's functions, third-order contracted kernels); (iv) the exact normalisation bookkeeping and the harmonic-space machinery needed to turn direction-space kernels into $C_\ell^{(22)}$ and $C_\ell^{(13)}$. Sections that assemble the explicit second- and third-order maps, the final $C_\ell$, the checks and the monochromatic reduction are marked as placeholders to be supplied. Every result quoted here has been verified symbolically or numerically; the verification ledger lists each check.

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# 1 Setup and conventions

**Model.** Spatially flat FRW background, general relativity with $\Lambda = 0$, signature $(-+++)$, and a single perfect radiation fluid

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + P g^{\mu\nu}, \qquad P = \rho/3, \qquad \rho = \alpha\, T_{\rm phys}^4, \tag{1}$$

with $\alpha$ constant and $u^\mu u_\mu = -1$. The second relation *defines* the physical temperature $T_{\rm phys}$; no chemical potential or particle-number assumption is made. Test photons are emitted isotropically with a Planck spectrum of temperature $T_*$ on the hypersurface $T_{\rm phys} = T_*$, then propagate collisionlessly without backreaction.

**Background and units.** Conformal time $\eta$, $\mathrm{d}s^2 = a^2(\eta)\,\hat{g}_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu$. In radiation domination $a \propto \eta$; we fix the normalisation $a(\eta) = \eta$, so that $\mathcal{H} \equiv a'/a = 1/\eta$ and the Friedmann equation $3\mathcal{H}^2 = 8\pi G a^2 \bar{\rho}$ reads

$$8\pi G\, a^4 \bar{\rho} = 3, \qquad \text{i.e.} \qquad 8\pi G\, \bar{\rho} = 3/\eta^4. \tag{2}$$

Background temperature $\bar{T}(\eta) = \bar{T}_i\, \eta_i/\eta$. The *conformal* fluid variables

$$\hat{\rho} \equiv a^4\rho, \qquad \hat{P} \equiv a^4 P, \qquad \hat{T} \equiv a\, T_{\rm phys}, \qquad \hat{u}^\mu \equiv a\, u^\mu \quad (\hat{g}_{\mu\nu}\hat{u}^\mu\hat{u}^\nu = -1), \tag{3}$$

are constant in the background ($\hat{\bar{\rho}} = 3/8\pi G$, $\hat{\bar{T}} = \text{const}$) and are the natural variables of Sec. 2.

**Fourier conventions and statistics.**

$$f(\mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, e^{i\mathbf{k}\cdot\mathbf{x}} f_{\mathbf{k}}, \qquad \langle \zeta_{\mathbf{k}}\zeta_{\mathbf{k}'}\rangle = (2\pi)^3\delta^3(\mathbf{k}+\mathbf{k}')\, P(k), \qquad \langle \zeta^2(\mathbf{x})\rangle = \int \frac{\mathrm{d}^3k}{(2\pi)^3} P(k) = \frac{1}{2\pi^2}\int k^2 \mathrm{d}k\, P(k). \tag{4}$$

Products in position space become convolutions, $(fg)_{\mathbf{k}} = \int \mathrm{d}^3q/(2\pi)^3\, f_{\mathbf{q}}\, g_{\mathbf{k}-\mathbf{q}}$, and $\partial_i \to ik_i$, $\nabla^2 \to -k^2$, $\nabla^{-2} \to -1/k^2$. $\langle\cdot\rangle$ denotes the ensemble average over the Gaussian field $\zeta_1$ (Sec. 3); $\langle\cdot\rangle_{\rm sky}$ denotes the solid-angle average over the observer's sky.

**Perturbative expansion.** Every perturbation $X \in \{\Psi, \Phi, \omega_i, h_{ij}, \delta, v^i, \ldots\}$ is expanded as

$$X = \epsilon X_1 + \frac{\epsilon^2}{2} X_2 + \frac{\epsilon^3}{6} X_3 + O(\epsilon^4), \tag{5}$$

and the same convention is used for every derived quantity ($S$, $\Lambda$, $m$, $\Theta$, $\zeta_{\rm nl}$). For an equation $E[X] = 0$ we write $E^{(n)} \equiv n!\,[\epsilon^n]E$, so that $X_n$ enters $E^{(n)}$ linearly with unit weight. The bookkeeping parameter $\epsilon$ is the amplitude of $\zeta_1$; the final expansion parameter is $\epsilon^2\langle\zeta_1^2\rangle$. Products of lower-order quantities are never dropped.

**Poisson gauge.** The conformal metric is parametrised *linearly* (no exponentials),

$$\hat{g}_{00} = -(1+2\Psi), \qquad \hat{g}_{0i} = \omega_i, \qquad \hat{g}_{ij} = (1-2\Phi)\delta_{ij} + h_{ij}, \qquad \partial_i\omega_i = 0, \quad h_{ii} = 0 = \partial_i h_{ij}, \tag{6}$$

with $\omega_i$ and $h_{ij}$ starting at order $\epsilon^2$ (no primordial vector or tensor modes, Sec. 3). Indices on $\omega_i$, $h_{ij}$, $v^i$ are raised and lowered with $\delta_{ij}$. The fluid is parametrised by

$$\rho = \bar{\rho}(\eta)\,(1+\delta), \qquad 1+\delta_T \equiv (1+\delta)^{1/4} = \hat{T}/\hat{\bar{T}}, \qquad \hat{u}^\mu = \hat{u}^0\,(1, v^i), \tag{7}$$

with $\hat{u}^0$ fixed by the normalisation $\hat{g}_{\mu\nu}\hat{u}^\mu\hat{u}^\nu = -1$ as a series in $\epsilon$. Irrotationality of $v^i$ is *not* imposed (Sec. 4.2). The Einstein equations are $E_{\mu\nu} \equiv G_{\mu\nu}[g] - 8\pi G\, T_{\mu\nu} = 0$ with $g = \eta^2\hat{g}$ and $8\pi G\, T_{\mu\nu} = (3/\eta^4)(1+\delta)[\frac{4}{3}u_\mu u_\nu + \frac{1}{3}g_{\mu\nu}]$; the conservation laws $\nabla_\mu T^{\mu\nu} = 0$ are implied and are used as checks. Dictionary to the exponential parametrisation of [8, 9, 6] ($\hat{g}_{00} = -e^{2\Phi_{\rm e}}$, $\hat{g}_{ij} = e^{-2\Psi_{\rm e}}\delta_{ij} + \ldots$, note the swapped names): $\Phi_{\rm e}^{(2)} = \Psi_2 - 2\Psi_1^2$, $\Psi_{\rm e}^{(2)} = \Phi_2 + 2\Phi_1^2$ at second order.

**Directions, rays and times.** $\hat{\mathbf{n}}$ is the direction in which the observer looks (unit vector with respect to the observer's orthonormal triad, Sec. 2.4); the photon propagation direction at the observer is $\hat{\mathbf{e}} = -\hat{\mathbf{n}}$. The background ray through the observer event $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$ is

$$\bar{\mathbf{x}}(\eta) = \bar{\mathbf{x}}_o + \hat{\mathbf{n}}\,\chi(\eta), \qquad \chi(\eta) \equiv \bar{\eta}_o - \eta, \qquad \frac{\mathrm{d}\bar{\mathbf{x}}}{\mathrm{d}\eta} = -\hat{\mathbf{n}}, \tag{8}$$

and the total derivative along it acting on a field is $D \equiv \partial_\eta - \hat{\mathbf{n}} \cdot \nabla$. Four times are kept distinct: the *initial* time $\eta_i$ at which initial data are posed ($\bar{T}(\eta_i) = T_i$); the background *emission* time $\eta_*$ defined by $\bar{T}(\eta_*) = T_*$, and the actual emission time $\eta_e(\hat{\mathbf{n}}) = \eta_* + \delta\eta_e(\hat{\mathbf{n}})$ along each ray; the *observation* time $\eta_o = \bar{\eta}_o + \delta\eta_o$ of the observer (direction independent, fixed by proper time); and the dummy time $\eta$ along the ray. The ordering is $0 < \eta_i < \eta_* < \bar{\eta}_o$. The angular quantities are

$$a_{\ell m}^{(r)} = \int \mathrm{d}\Omega_{\hat{\mathbf{n}}}\, Y_{\ell m}^*(\hat{\mathbf{n}})\, \Theta^{(r)}(\hat{\mathbf{n}}), \qquad C_\ell^{(rs)} = \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} \langle a_{\ell m}^{(r)} a_{\ell m}^{(s)*} \rangle_c, \tag{9}$$

with $Y_{\ell m}$ carrying the Condon–Shortley phase, and the target is

$$C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4 \Big[ \frac{1}{4} C_\ell^{(22)} + \frac{1}{3} \operatorname{Re} C_\ell^{(13)} \Big] + O(\epsilon^6), \qquad \ell \geq 2. \tag{10}$$

**Relation to existing work.** No complete $O(\epsilon^4)$ (one-loop) computation of the temperature power spectrum exists in the literature: the only (22)-type temperature calculation [23] is for dust in synchronous gauge (ISW terms only) and explicitly leaves out the (13) term for lack of third-order metric perturbations; third-order temperature anisotropies have been studied on large scales in matter domination [24, 25]; second-order Boltzmann codes [20, 22] and the (22)-type $B$-mode spectrum [21] exist, as do the second-order Sachs–Wolfe analyses [18, 19, 8, 9, 10] which, "following standard practice", drop the observer terms; only [6, 7] keep them, for long-wavelength modes. The lensed $C_\ell$ at $O(C^\phi)$ [16, 17] is the (22) + (13) structure restricted to lensing and serves as our control case. The radiation-era second-order vector and tensor sectors are treated in [15] and [12, 13, 14]; general second-order perturbation theory is reviewed in [11].

# 2 The exact observable

## 2.1 Conformal frame

Write $g_{\mu\nu} = a^2 \hat{g}_{\mu\nu}$ with $a = \eta$. Null geodesics of $g$ and $\hat{g}$ coincide as curves; if $p^\mu = \mathrm{d}x^\mu/\mathrm{d}\lambda$ is affine for $g$ then $\hat{p}^\mu = a^2 p^\mu$ is affine for $\hat{g}$. With the conformal fluid variables (3),

$$E \equiv -u_\mu p^\mu = \hat{E}/a, \qquad \hat{E} \equiv -\hat{g}_{\mu\nu} \hat{u}^\mu \hat{p}^\nu. \tag{11}$$

The radiation stress tensor is traceless, hence conformally covariant: $\hat{T}^{\mu\nu} \equiv a^6 T^{\mu\nu} = (\hat{\rho} + \hat{P}) \hat{u}^\mu \hat{u}^\nu + \hat{P} \hat{g}^{\mu\nu}$ with $\hat{\rho} = \alpha \hat{T}^4$, $\hat{P} = \hat{\rho}/3$, and $\hat{\nabla}_\mu \hat{T}^{\mu\nu} = 0$ *exactly*, where $\hat{\nabla}$ is the Levi-Civita connection of $\hat{g}$. Everything in this section therefore holds verbatim for the hatted system, whose background is Minkowski space with a constant fluid; we drop the hats in the lemmas (they hold for any metric) and restore them when we parametrise.

## 2.2 Two lemmas

Let $u^\mu$ be a unit timelike field and decompose its gradient in the standard way,

$$\nabla_\nu u_\mu = \sigma_{\mu\nu} + \omega_{\mu\nu} + \frac{1}{3}\theta\, h_{\mu\nu} - u_\nu a_\mu, \qquad h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu, \quad a_\mu = u^\nu \nabla_\nu u_\mu, \quad \theta = \nabla_\mu u^\mu, \tag{12}$$

with $\sigma_{\mu\nu}$ the symmetric trace-free part and $\omega_{\mu\nu}$ the antisymmetric part of $h_\mu{}^\alpha h_\nu{}^\beta \nabla_\beta u_\alpha$ [2]. For a null vector $p^\mu$ write $p^\mu = E(u^\mu + e^\mu)$ with $E = -u\cdot p$, $u\cdot e = 0$, $e\cdot e = 1$, and define the proper-length element $\mathrm{d}l \equiv E\,\mathrm{d}\lambda$ along the ray.

**Lemma 2.1** (kinematic redshift identity)**.** *Along an affinely parametrised null geodesic, $p^\nu \nabla_\nu p^\mu = 0$,*

$$\frac{\mathrm{d}\ln E}{\mathrm{d}l} = -\Big(\frac{\theta}{3} + \sigma_{\mu\nu} e^\mu e^\nu + a_\mu e^\mu\Big). \tag{13}$$

*Proof.* $\mathrm{d}E/\mathrm{d}\lambda = p^\nu \nabla_\nu(-u_\mu p^\mu) = -p^\mu p^\nu \nabla_\nu u_\mu$. Insert (12): the $\sigma$ term gives $E^2\sigma_{ee}$ ($\sigma \perp u$); the $\omega$ term vanishes identically (antisymmetric tensor contracted with $p^\mu p^\nu$), so *vorticity drops out with no assumption*; the $\theta$ term gives $\frac{1}{3}\theta E^2 h_{ee} = \frac{1}{3}\theta E^2$; the acceleration term gives $-E^2[u\cdot(u+e)][a\cdot(u+e)] = E^2 a\cdot e$. Divide by $E^2$. □

**Lemma 2.2** (radiation-fluid identities)**.** *For the fluid (1), the conservation law $\nabla_\mu T^{\mu\nu} = 0$ alone (no field equations) is equivalent to*

$$\theta = -3\, u^\mu \partial_\mu \ln T, \qquad a_\mu = -h_\mu{}^\nu \partial_\nu \ln T. \tag{14}$$

*Proof.* $u_\nu \nabla_\mu T^{\mu\nu} = 0$ gives $u\cdot\partial\rho + (\rho+P)\theta = 0$ and $h^\nu{}_\lambda \nabla_\mu T^{\mu\lambda} = 0$ gives $(\rho+P)a^\nu + h^{\nu\mu}\partial_\mu P = 0$. With $\rho = \alpha T^4$, $P = \rho/3$ one has $\rho + P = \frac{4}{3}\alpha T^4 = T\,\mathrm{d}P/\mathrm{d}T$ and $\mathrm{d}\rho/\mathrm{d}T = 3(\rho+P)/T$, which turn the two equations into (14). Both ingredients are needed: for $P = w\rho$, $\rho \propto T^4$ one finds $a = -\frac{4w}{1+w} h\,\partial \ln T$ and $\theta = -\frac{4}{1+w} u\cdot\partial \ln T$, which reduce to (14) iff $w = \frac{1}{3}$ (App. B, V1-2). □

## 2.3 The theorem

**Theorem 2.3.** *Let the photon be emitted at the event $e$ where its past-directed null geodesic last meets the hypersurface $T = T_*$, with an isotropic Planck spectrum in the fluid rest frame, and let it be received at the event $o$ by an observer comoving with the fluid. Then the observed temperature in the direction $\hat{\mathbf{n}}$ is*

$$\boxed{T_{\mathrm{obs}}(\hat{\mathbf{n}}) = T_* \frac{E_o}{E_e} = T_o \exp[-S(\hat{\mathbf{n}})], \qquad S(\hat{\mathbf{n}}) \equiv \int_e^o \sigma_{\mu\nu} e^\mu e^\nu \,\mathrm{d}l} \tag{15}$$

*where $T_o$ is the fluid temperature at $o$ and the integral runs along the ray.*

*Proof.* Since $e^\mu h_\mu{}^\nu = e^\nu$, Lemma 2.2 gives $\theta/3 + a\cdot e = -(u+e)\cdot\partial \ln T = -E^{-1} p\cdot\partial \ln T = -\mathrm{d}\ln T/\mathrm{d}l$. Lemma 2.1 then reads $\mathrm{d}(\ln E - \ln T)/\mathrm{d}l = -\sigma_{ee}$; $p$ is future directed, so $\lambda$ increases from $e$ to $o$ and $\ln(E_o/E_e) = \ln(T_o/T_e) - S$. With $T_e = T_*$ this is $T_* E_o/E_e = T_o e^{-S}$. Finally, $E_o/E_e$ is independent of the photon energy (the geodesic equation is homogeneous in $p$), so by Liouville's theorem a Planck spectrum at $T_*$ in the fluid frame at $e$ is received as a Planck spectrum at $T_* E_o/E_e$ in the fluid frame at $o$; this is what $T_{\mathrm{obs}}(\hat{\mathbf{n}})$ means. □

*Remark* 2.4 (conformal invariance). Theorem 2.3 applies to $(g, u, T)$ and to $(\hat{g}, \hat{u}, \hat{T})$ alike. Since $E = \hat{E}/a$ and $T = \hat{T}/a$ with the same $a_e$, $a_o$, the two statements agree iff $S = \hat{S}$: the shear integral is conformally invariant and may be computed entirely in the hatted frame, where the background shear vanishes identically.

**Corollary 2.5.** *With* $\Theta(\hat{\mathbf{n}}) \equiv T_{\rm obs}(\hat{\mathbf{n}})/\langle T_{\rm obs}\rangle_{\rm sky} - 1$ *and* $\langle\cdot\rangle_{\rm sky} = \dfrac{1}{4\pi}\int \mathrm{d}\Omega$ *in the observer's rest frame,*

$$1 + \Theta(\hat{\mathbf{n}}) = \frac{e^{-S(\hat{\mathbf{n}})}}{\langle e^{-S}\rangle_{\rm sky}} \qquad \textit{exactly.} \tag{16}$$

**Consequences.**

(i) All of the usual emission-surface (intrinsic temperature and $T_{\rm phys} = T_*$ shift), Sachs–Wolfe, Doppler, integrated Sachs–Wolfe, lensing and time-delay physics is contained in $S$: the only inputs are the fluid shear field $\sigma_{\mu\nu}$, the ray $x^\mu(\lambda)$, and the two endpoints. In particular the perturbation of the emission point along the ray does *not* require a separate "intrinsic $\delta T$" term; $T_e = T_*$ has been used exactly.

(ii) $T_o$ is a scalar at the observer event, independent of $\hat{\mathbf{n}}$; it drops out of $\Theta$ identically. No expansion of $T_o$ or of $\langle T_{\rm obs}\rangle_{\rm sky}$ in terms of a background temperature is ever needed.

(iii) The observer's tetrad. Fermi–Walker transport of an orthonormal tetrad along the observer's worldline defines the sky directions. The FW tetrad is a realisation-dependent rigid rotation $R = \exp(\epsilon A)$ of the *canonical triad* (Sec. 2.4). A rigid rotation of the sky leaves $\sum_m |a_{\ell m}|^2$ invariant realisation by realisation, hence $C_\ell$ is independent of the tetrad orientation. However, because $R$ depends on $\epsilon$, $a^{(2)}_{\ell m} \to a^{(2)}_{\ell m} + 2A\, a^{(1)}_{\ell m}$ etc., so $C^{(22)}_\ell$ and $\mathrm{Re}\, C^{(13)}_\ell$ are *not separately* invariant; only $\dfrac{1}{4}C^{(22)}_\ell + \dfrac{1}{3}\,\mathrm{Re}\, C^{(13)}_\ell$ is (verified explicitly at $\ell = 2$, App. B, V1-5). Consequently $\Theta^{(1)}, \Theta^{(2)}, \Theta^{(3)}$ must all be computed in one and the same frame; we use the canonical triad throughout. The dipole power $\sum_m |a_{1m}|^2$ is likewise invariant, so "remove the dipole after normalising" is frame independent.

(iv) The emission surface enters only through the endpoint $e$ of the integral: the location of the last crossing of $T = T_*$ along the past ray. Perturbatively there is a single crossing and $\eta_e(\hat{\mathbf{n}}) = \eta_* + \delta\eta_e(\hat{\mathbf{n}})$.

*Remark* 2.6 (validity). The proof uses the differential form of $\nabla \cdot T = 0$ and $C^1$ fields along the ray; it is not defined across a shock (irrelevant perturbatively). Isotropic Planck emission in the fluid frame at $T = T_*$ is self-consistent (the photons *are* the fluid there); after emission the "fluid" $(u, T)$ entering $S$ is the model's perfect-fluid continuation, as the problem statement prescribes. The identities also hold for $\rho = 3P + c$, $P = CT^4 - c/4$ (a $\Lambda$-like constant), not needed here.

## 2.4 Observer, clock, emission event and direction map

*Observer worldline*: a tagged fluid element, $\mathbf{x}_o(\eta) = \mathbf{x}^{\rm lab}_o + \int_{\eta_{\rm init}}^{\eta} v^i(\eta', \mathbf{x}_o(\eta'))\,\mathrm{d}\eta'$. *Clock*: proper time elapsed since the observer crossed the initial-data surface $T_{\rm phys} = T_i$ (Poisson time $\eta_{\rm init}(\mathbf{x}_o)$ with $1 + \delta_T(\eta_{\rm init}, \mathbf{x}_o) = \eta_{\rm init}/\eta_i$, since $\bar{T} \propto 1/\eta$),

$$\Delta\tau = \int_{\eta_{\rm init}}^{\eta_o} \frac{a(\eta)}{\hat{u}^0(\eta, \mathbf{x}_o(\eta))}\,\mathrm{d}\eta \quad \text{held fixed} \qquad \Longrightarrow \qquad \eta_o = \bar{\eta}_o + \delta\eta_o[\zeta], \tag{17}$$

with $\delta\eta_o$ direction independent (it is a functional of the fields on the worldline). *Emission event*: the ray meets $T_{\rm phys} = T_*$, i.e.

$$1 + \delta_T(\eta_e, \mathbf{x}(\eta_e)) = \eta_e/\eta_*. \tag{18}$$

*Canonical triad*: $\{e_a\}_{a=1,2,3}$ is the Gram–Schmidt orthonormalisation (order $1,2,3$) of the coordinate basis $\partial_i$ with respect to $\hat{g}$, orthogonal to $\hat{u}$. *Direction map*: the photon received from the sky direction $\hat{\mathbf{n}}$ has

$$\hat{p}_o = \hat{E}_o(\hat{u}_o - n^a e_a), \qquad k^i(\eta_o) \equiv \frac{\mathrm{d}x^i}{\mathrm{d}\eta}\Big|_o = \frac{\hat{p}^i_o}{\hat{p}^0_o}, \tag{19}$$

so that in the background $k^i = -n^i$ and the ray is (8). With the ray parametrised by $\eta$, $k^\mu = \mathrm{d}x^\mu/\mathrm{d}\eta = (1, k^i)$, and using $\sigma \perp u$, $\hat{E}/\hat{p}^0 = -\hat{u}_\mu k^\mu$,

$$S(\hat{\mathbf{n}}) = \int_{\eta_e(\hat{\mathbf{n}})}^{\eta_o} \mathrm{d}\eta \, \frac{\hat{\sigma}_{\mu\nu}k^\mu k^\nu}{-\hat{u}_\mu k^\mu}\Big|_{x^i(\eta)}. \tag{20}$$

The perturbative evaluation of (20) (ray, endpoints, triad, clock, all to third order) is the subject of Secs. 6–7.

### 2.5 Linear order: recovery of the Sachs–Wolfe formula

At $O(\epsilon)$ the endpoints and the ray may be taken at their background values, since $S$ itself starts at $O(\epsilon)$. With $\hat{g} = \mathrm{diag}(-(1+2\epsilon\Psi), (1-2\epsilon\Phi)\delta_{ij})$, $\hat{u}^\mu = (1-\epsilon\Psi, \epsilon v^i)$, $\hat{T} = \bar{\hat{T}}(1+\epsilon\delta_T)$, the covariant definitions (12) give (App. B, V1-4)

$$\hat{\theta} = \epsilon(\partial\cdot v - 3\Phi'), \qquad \hat{a}_i = \epsilon(v_i' + \partial_i\Psi), \qquad \hat{\sigma}_{ij} = \epsilon[\partial_{(i}v_{j)} - \frac{1}{3}\delta_{ij}\partial\cdot v], \qquad \hat{\sigma}_{0\mu} = 0, \tag{21}$$

and Lemma 2.2 becomes the linearised continuity and Euler equations

$$\delta_T' = \Phi' - \frac{1}{3}\partial\cdot v, \qquad v_i' = -\partial_i(\Psi + \delta_T). \tag{22}$$

With $e = -\hat{\mathbf{n}}$ and $D = \partial_\eta - \hat{\mathbf{n}}\cdot\nabla$ along the background ray one has $n^i n^j \partial_i v_j = \hat{\mathbf{n}}\cdot v' - D(\hat{\mathbf{n}}\cdot v)$, $\hat{\mathbf{n}}\cdot\nabla f = f' - Df$; inserting (22),

$$\hat{\sigma}_{1,ee} = n^i n^j \partial_i v_j - \frac{1}{3}\partial\cdot v = D(\delta_T + \Psi - \hat{\mathbf{n}}\cdot v) - (\Phi' + \Psi') \qquad \text{(on shell)}. \tag{23}$$

Integrating along the ray from $(\eta_*, \bar{\mathbf{x}}(\eta_*))$ to $(\bar{\eta}_o, \bar{\mathbf{x}}_o)$,

$$\boxed{L_1 \equiv -S_1 = [\delta_T + \Psi - \hat{\mathbf{n}}\cdot v]_e - [\delta_T + \Psi - \hat{\mathbf{n}}\cdot v]_o + \int_{\eta_*}^{\bar{\eta}_o} (\Phi' + \Psi')\,\mathrm{d}\eta} \tag{24}$$

which is the standard Sachs–Wolfe [1] + Doppler + integrated Sachs–Wolfe formula with the emission-surface shift (the $\delta_T$ at $e$ arises because $\bar{T} \propto 1/\eta$ and the emission surface is $T_{\rm phys} = T_*$, not $\eta = \eta_*$) and with the observer terms: a monopole $-[\delta_T + \Psi]_o$ and the observer dipole $+\hat{\mathbf{n}}\cdot v_o$. Both are removed from $\Theta^{(1)}$ by the normalisation and the dipole subtraction, but they multiply other terms at higher order and are kept (Sec. 5.1). Equation (23) has been verified symbolically and the same identity holds through third order at the integrand level (App. B, C2-1).

# 3 Primordial variable and initial data

## 3.1 The non-linear primordial variable

**Definition 3.1.** In Poisson gauge, with the non-linear $\delta_T$ and $\Phi$ of (7), (6),

$$\zeta_{\rm nl}(\mathbf{x}) \equiv \lim_{\eta\to 0}\Big[\ln\left(1+\delta_T(\eta,\mathbf{x})\right)+\frac{1}{2}\ln\left(1-2\Phi(\eta,\mathbf{x})\right)\Big] = \lim_{\eta\to 0}\Big[\frac{1}{4}\ln(1+\delta)+\frac{1}{2}\ln(1-2\Phi)\Big]. \quad (25)$$

*Why this is the uniform-density curvature perturbation.* Since $\rho=\bar\rho(\eta)(1+\delta)$ with $\bar\rho\propto\eta^{-4}$, the uniform-density hypersurface $\rho=\bar\rho(\tilde\eta)$ is the surface of Poisson time $\eta_P(\mathbf{x})=\tilde\eta\,[1+\delta_T(\eta_P,\mathbf{x})]$. Its induced metric is $\gamma_{ij}=g_{ij}+g_{00}\partial_i\eta_P\partial_j\eta_P+2g_{0(i}\partial_{j)}\eta_P$ with $g=\eta_P^2\hat g$, i.e.

$$\gamma_{ij}=\tilde\eta^2(1+\delta_T)^2\Big[(1-2\Phi)\delta_{ij}+h_{ij}+O(\tilde\eta^2)\Big], \quad (26)$$

because $\partial_i\eta_P=\tilde\eta\,\partial_i\delta_T=O(\tilde\eta)$. For regular solutions (below) $h_{ij}\to 0$ as $\tilde\eta\to 0$ (the induced tensors are sourced and vanish at $\eta=0$; there are no free tensor modes), so $\gamma_{ij}\to a(\tilde\eta)^2e^{2\zeta_{\rm nl}}\delta_{ij}$: $\zeta_{\rm nl}$ is exactly the quantity $\zeta$ of Salopek–Bond and Maldacena, $h_{ij}=a^2e^{2\zeta}(e^\gamma)_{ij}$ on uniform-density slices with $\gamma=0$ [3, 4]. (On these slices the fluid is also comoving in the limit, since the regular velocity potential vanishes as $\eta\to 0$, Sec. 4.1.) The spatial coordinates are those of Poisson gauge; the residual freedom of the conformally flat form, a constant rescaling $\mathbf{x}\to B\mathbf{x}$, is the zero-mode ambiguity discussed in Sec. 3.4.

**Definition 3.2** (Gaussian initial conditions; absence of higher-order primordial modes)**.**

$$\zeta_{\rm nl}=\epsilon\,\zeta_1 \quad \text{exactly}, \qquad \zeta_1 \text{ a Gaussian random field with spectrum } P(k), \quad (27)$$

i.e. in the expansion (5) of $\zeta_{\rm nl}$, $\zeta^{(2)}(\eta\to 0)=0$ and $\zeta^{(3)}(\eta\to 0)=0$ *identically in* $\mathbf{x}$. This is the statement that the exponentiated uniform-density curvature perturbation is Gaussian [4]; in the language of [10] it is $\mathcal{R}^{(2)}=-2\mathcal{R}^{(1)2}$ ($f_{\rm NL}=0$ for the variable appearing in $e^{2\zeta}$), and it fixes the second- and third-order scalar fields at $\eta\to 0$ completely, with no freedom to add independent second- or third-order scalar, vector or tensor modes.

## 3.2 Regularity and initial data at finite $\eta_i$

*Regular* means: at every order the solution of the Einstein–fluid system is analytic in $\eta$ at $\eta=0$. At first order this discards the decaying mode $\propto(\cos x+x\sin x)/x^3\sim\eta^{-3}$ ($x=k\eta/\sqrt{3}$); at second and third order it discards the homogeneous scalar decaying modes and all homogeneous vector and tensor modes, and selects the particular solutions that vanish at $\eta=0$ (Sec. 4.2). The regular solution is uniquely fixed by (27).

Initial data at the finite time $\eta_i$ ($\bar T(\eta_i)=T_i$) are *defined* as the restriction of the regular solution to $\eta=\eta_i$. They satisfy the Hamiltonian and momentum constraints at every order by construction, since the regular solution solves all ten Einstein equations for $\eta>0$ and the constraints propagate (contracted Bianchi identity, verified to $O(\epsilon^3)$ by the engine, App. B, C1-5). Concretely, every Green's-function integral below may be written either as $\int_0^\eta$ or as $\int_{\eta_i}^\eta$ plus the homogeneous solution matched to the (regular) data at $\eta_i$; the two are identical. We keep $\eta_i$ explicit where the problem statement asks for it and note that $\eta_i\to 0$ reproduces the $\int_0^\eta$ form. This is how the instruction "do not indiscriminately set higher-order perturbations to zero" is implemented: $X_2(\eta_i)$, $X_3(\eta_i)$ are non-zero, non-local functionals of $\zeta_1$ determined by (27) and the constraints.

### 3.3 The conditions at second and third order

Expanding (25) with (5) (engine output, App. B, C1-8),

$$\zeta^{(1)} = \frac{1}{4}\delta_1 - \Phi_1 = \delta_{T,1} - \Phi_1, \tag{28}$$

$$\zeta^{(2)} = \frac{1}{4}\delta_2 - \frac{1}{4}\delta_1^2 - \Phi_2 - 2\Phi_1^2, \tag{29}$$

$$\zeta^{(3)} = \frac{1}{4}\delta_3 - \frac{3}{4}\delta_1\delta_2 + \frac{1}{2}\delta_1^3 - \Phi_3 - 6\Phi_1\Phi_2 - 8\Phi_1^3. \tag{30}$$

At $\eta \to 0$ the first-order regular solution has (Sec. 4.1) $\Phi_1 = \Psi_1 \to -\frac{2}{3}\zeta_1$, $\delta_1 \to \frac{4}{3}\zeta_1$, $v_1 \to 0$.

**Second order.** In the kernel notation of Sec. 4.2 ($X_{2,\mathbf{k}} = \int \frac{\mathrm{d}^3q}{(2\pi)^3} K_X(\mathbf{q}, \mathbf{k}-\mathbf{q}; \eta)\zeta_{\mathbf{q}}\zeta_{\mathbf{k}-\mathbf{q}}$, $\mathbf{p} \equiv \mathbf{k} - \mathbf{q}$, local products of first-order fields being constant kernels at $\eta \to 0$), the 00 constraint gives $\delta_2(0) = \frac{1}{3}\lim(\eta^2 S_{00}) - 2\Psi_2(0)$ with $\lim_{\eta\to 0}\eta^2 K_{S_{00}} = \frac{32}{3}$, the traceless $ij$ equation gives $\Psi_2(0) = \Phi_2(0) - K_Q(0)$ with

$$K_Q(\mathbf{q},\mathbf{p};0) = -\frac{13k^4 - 6k^2(p^2+q^2) + 9(p^2-q^2)^2}{9k^4} = -\frac{4}{9}\,\frac{4k^2 - 6kq\mu + 9q^2\mu^2 - 3q^2}{k^2}, \qquad \mu \equiv \hat{\mathbf{k}}\cdot\hat{\mathbf{q}}, \tag{31}$$

and $\zeta^{(2)}(0) = 0$ then yields

$$\boxed{K_{\Phi_2}(\mathbf{q},\mathbf{p};0) = \frac{1}{3}K_Q(\mathbf{q},\mathbf{p};0) - \frac{8}{27} = -\frac{7k^4 - 2k^2(p^2+q^2) + 3(p^2-q^2)^2}{9k^4} = -\frac{4}{9}\,\frac{2k^2 - 2kq\mu + 3q^2\mu^2 - q^2}{k^2}} \tag{32}$$

(the reader may verify: $\frac{1}{4}[\frac{32}{9} + 2K_Q(0) - 2\Phi_2] - \frac{1}{4}\cdot\frac{16}{9} - \Phi_2 - 2\cdot\frac{4}{9} = 0$). In position space, with $\Phi_1 = -\frac{2}{3}\zeta_1$,

$$\Phi_2(0) = -2\Phi_1^2 - \nabla^{-2}(\partial_i\Phi_1\partial_i\Phi_1) + 3\nabla^{-4}\partial_i\partial_j(\partial_i\Phi_1\partial_j\Phi_1), \tag{33}$$

$$\Psi_2(0) - \Phi_2(0) = 4\Phi_1^2 + 3\nabla^{-2}(\partial_i\Phi_1\partial_i\Phi_1) - 9\nabla^{-4}\partial_i\partial_j(\partial_i\Phi_1\partial_j\Phi_1), \tag{34}$$

and $\delta_2(0)$, $v_2(0)$ follow from the constraints (App. A). Equations (33)–(34) agree *exactly* with the radiation-era superhorizon initial conditions of Pitrou–Uzan–Bernardeau [10] (their (3.12b,c) with $\mathcal{R}^{(2)} = -2\mathcal{R}^{(1)2}$, after exchanging their names for the two potentials). Bartolo–Matarrese–Riotto [9] (their (7.30)) agree in the local and gradient terms but differ in the velocity-quadrupole term, which they evaluate with the late-time velocity rather than its $\eta \to 0$ limit (App. B, C1-11).

**Third order.** Only the $q$-contracted third-order fields are needed for $C_\ell^{(13)}$ (Sec. 4.3). With $\Phi_{3c}(k,\eta) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\, K_{\Phi_{3c}}(k,q,\mu;\eta)$, the regular solution has $\Phi_{3c}(k,0) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} P(q)\, C_3(k;q,\mu)$ and $\zeta^{(3)}(0) = 0$ gives

$$\boxed{C_3(k;q,\mu) = -\frac{16(2k^5 + 12k^4q\mu - 42k^3q^2\mu^2 - 10k^3q^2 + 36k^2q^3\mu^3 + 32k^2q^3\mu + 3kq^4\mu^4 - 42kq^4\mu^2 + kq^4 + 8q^5\mu)}{9\,k\,(k^2 - 2kq\mu + q^2)^2}} \tag{35}$$

($p^2 = k^2 - 2kq\mu + q^2$; invariant form and the intermediate $Q_{3c}(0)$, $\lim \eta^2 K_{3c,S_{00}}$ in App. A). The $p^{-4}$ denominator is the Poisson inversion of the traceless equation; it is integrable over $\mu$ at $q \to k$ (the numerator vanishes there to the required order).

### 3.4 The exact dilation family and the zero-mode (tadpole) ambiguity

The flat FRW solution written in rescaled coordinates $\tilde{\eta} = A\eta$, $\tilde{\mathbf{x}} = B\mathbf{x}$ ($A, B$ constants) is, in the $(\eta, \mathbf{x})$ chart, the exact Poisson-gauge configuration

$$1 + 2\Psi = A^4, \qquad 1 - 2\Phi = A^2 B^2, \qquad 1 + \delta = A^{-4}, \qquad v^i = 0, \qquad \zeta_{\rm nl} = \ln B, \tag{36}$$

which the engine verifies to solve $E_{\mu\nu} = 0$ through $O(\epsilon^3)$ for arbitrary $A, B$ (App. B, C1-6). It is the $k = 0$ member of the solution space: a constant $\zeta_1 = c$ corresponds to $B = e^{\epsilon c}$ (so that $\zeta_{\rm nl} = \epsilon c$ exactly) and, by (28), $a_1 = -c/3$, giving $\Psi_1 = \Phi_1 = -\frac{2}{3}c$, $\delta_1 = \frac{4}{3}c$ in agreement with the $k \to 0$ limit of the first-order transfer functions. At second order, however, the $k = 0$ equations fix only $\delta_2 = -2\Psi_2 + \frac{32}{9}c^2$ and leave $\Phi_2, \Psi_2$ free: this is the residual gauge freedom of Poisson gauge at zero momentum (the time rescaling $A$, cf. Weinberg's adiabatic modes [5, 6]). Correspondingly, the $k \to 0$ limit of the finite-$k$ kernels (31)–(32) at $\mathbf{p} = -\mathbf{q}$ is direction dependent ($\propto 3\mu^2 - 1$) and its angular average,

$$\langle \Phi_2(0) \rangle_{\rm ang} = -\frac{8}{9}c^2, \qquad \langle \Psi_2(0) \rangle_{\rm ang} = +\frac{8}{9}c^2, \qquad \delta_2 = \frac{16}{9}c^2, \tag{37}$$

lies on the family (36) with $A = e^{-\epsilon c/3}$, $B = e^{\epsilon c}$, i.e. $1 + \delta_T = e^{\zeta/3}$, $1 - 2\Phi = e^{4\zeta/3}$, $1 + 2\Psi = e^{-4\zeta/3}$ (the member with $\Psi_2 = \Phi_2$ instead has $\Phi_2 = \Psi_2 = -\frac{8}{27}c^2$, $\delta_2 = \frac{112}{27}c^2$).

This matters because the third-order map contains *mean-field* (tadpole) terms $\langle X_2 \rangle = \int \frac{{\rm d}^3 q}{(2\pi)^3} P(q) K_X(\mathbf{q}, -\mathbf{q}; \eta)$, the $\mathbf{k} = 0$ components of the second-order fields (Sec. 4.3). They are evaluated with the angle-averaged convention (closed forms ${\rm tad}_X(q, \eta)$ in App. A; e.g. ${\rm tad}_{\Phi_2} \to -\frac{8}{9}$, ${\rm tad}_{Q_2} \to -\frac{16}{9}$, ${\rm tad}_{\delta_2} \to \frac{16}{9}$ as $\eta \to 0$). Since the whole $k = 0$ family is a coordinate transformation and the observable is defined by physical labels only (fluid worldline, proper time from the $T_{\rm phys} = T_i$ crossing, emission on $T_{\rm phys} = T_*$, sky-monopole normalisation), a constant $\zeta_1$ must be unobservable: the full map must be invariant under $c$ at every order, which fixes the squeezed limits of $K_2$ and $K_3$ and forces the tadpole convention to drop out of $\Theta^{(3)}$ and hence of $C_\ell^{(13)}$. This is the dilation check, to be performed on the assembled map (Sec. 9); at the level of the field equations the consistency is already verified by the squeezed limit of (35),

$$\lim_{q \to 0} C_3(k; q, \mu) = -\frac{32}{9}, \tag{38}$$

which is precisely the $c^2 \zeta_s$ coefficient $6 \times \frac{1}{2}(\frac{4}{3}c)^2 \times (-\frac{2}{3})$ of the short mode modulated by the long-mode member $A = e^{-\epsilon c/3}$, $B = e^{\epsilon c}$ ($1 - 2\Phi = A^2 B^2 (1 - 2\Phi_s)$, App. B, C1-14).

## 4 Perturbative solution of the Einstein–fluid equations

This section summarises the solution of $E_{\mu\nu} = G_{\mu\nu}[\eta^2 \hat{g}] - 8\pi G T_{\mu\nu} = 0$ through $O(\epsilon^3)$ in the parametrisation of Sec. 1, obtained with a dedicated truncated-polynomial engine (code `cmb4/code/C1_einstein/`, App. B, C1-$*$). The Christoffel symbols, Ricci and Einstein tensors are computed directly from $g = \eta^2 \hat{g}$ as series in $\epsilon$; the background part of every component vanishes identically and the contracted Bianchi identity $g^{\mu\alpha} \nabla_\alpha E_{\mu\nu} + g_{\nu\beta} \nabla_\mu T^{\mu\beta} = 0$ holds identically through $O(\epsilon^3)$. Table 1 collects the named objects used in the rest of the paper; each is defined once in the text and the long expressions are in App. A.

| object | defined in | meaning |
|---|---|---|
| $x = k\eta/\sqrt{3}$ | (40) | dimensionless sound-horizon variable |
| $T_\Phi, T_\delta, T_{\delta_T}, T_V$ | (41) | first-order transfer functions ($X_1 = \zeta_1 T_X$) |
| $\mathcal{L}_\Phi, \mathcal{L}_h, \mathcal{L}_\omega$ | (44) | universal linear operators (same at every order) |
| $S^{(n)}_{\mu\nu}$; $S_{\rm TL}, S^L_0, S^T_{0i}, S^V_{ij}, S^{TT}_{ij}, S_\Phi$ | (42), (43) | order-$n$ sources and their SVT projections |
| $K_X(\mathbf{q},\mathbf{p};\eta)$, $\mathbf{p} = \mathbf{k} - \mathbf{q}$ | (46) | second-order kernel of $X$ per $\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}$ |
| $K_Q, K_{S_{00}}, K_{S_{0z}}, K_{S_\Phi}, K_\omega, K_{v^T}, K_h, K^0_{\Phi_2}$ | (47), App. A | the explicit second-order kernels |
| $u_1, u_2, \hat{g}, G_\Phi, G_h$ | (48) | homogeneous solutions and Green's functions |
| $\mathcal{K}_{\Phi_2}, \mathcal{K}_h$ | (49) | declared time integrals for $\Phi_2$, $h_2$ |
| $X_{3c}(k,\eta)$, $K_{3c,X}(k,q,\mu;\eta)$ | (50) | $q$-contracted third-order fields and kernels |
| $\mathrm{tad}_X(q,\eta)$ | (52) | mean-field (tadpole) functions $\langle X_2\rangle$ |
| $C_3(k;q,\mu)$ | (35) | third-order superhorizon amplitude of $\Phi_{3c}$ |

Table 1: Named objects of the perturbative solution.

### 4.1 First order

In the frame $\mathbf{k} = k\hat{\mathbf{z}}$ the first-order equations are

$$\begin{aligned}
E^{(1)}_{ij}\ (i\neq j,\ \text{traceless}): &\quad -k^2(\Phi_1-\Psi_1) = 0 \ \Rightarrow\ \Psi_1 = \Phi_1,\\
E^{(1)}_{0i}: &\quad 2ik_i(\Phi_1' + \Psi_1/\eta) + 4v_{1,i}/\eta^2 = 0 \ \Rightarrow\ v_1^T = 0, \quad v_{1,i} = \partial_i V_1,\ V_1 = -\frac{\eta^2}{2}(\Phi_1' + \Psi_1/\eta),\\
E^{(1)}_{00}: &\quad \delta_1 = -\frac{2}{3}\eta^2 k^2\Phi_1 - 2\eta\Phi_1' - 2\Psi_1,\\
\text{trace}: &\quad \Phi_1'' + \frac{4}{\eta}\Phi_1' + \frac{k^2}{3}\Phi_1 = 0.
\end{aligned} \tag{39}$$

The velocity is irrotational and the conservation equations hold identically on this solution (App. B, C1-3). With

$$x \equiv k\eta/\sqrt{3} \tag{40}$$

the regular solution, normalised by $\zeta_1 = \lim_{\eta\to 0}(\delta_{T,1} - \Phi_1)$, is $X_1 = \zeta_1 T_X(x)$ with

$$\begin{aligned}
T_\Phi(x) &= -\frac{2(\sin x - x\cos x)}{x^3} \ \xrightarrow{x\to 0}\ -\frac{2}{3}, \qquad \Psi_1 = \Phi_1,\\
T_\delta(x) &= 4T_{\delta_T}(x) = -4\cos x + \frac{8\sin x}{x} + \frac{8\cos x}{x^2} - \frac{8\sin x}{x^3} \ \to\ \frac{4}{3},\\
T_V(x) &= \eta\,\frac{x^2\sin x + 2x\cos x - 2\sin x}{x^3} \ \to\ \frac{\eta}{3} \qquad (v_{1,i} = \partial_i V_1,\ V_1 = \zeta_1 T_V).
\end{aligned} \tag{41}$$

The second solution $(\cos x + x\sin x)/x^3 \sim \eta^{-3}$ is discarded by regularity. Note $\delta_{T,1} \to \zeta_1/3$, so $\zeta_1 = \delta_{T,1} - \Phi_1 = \frac{1}{3}\zeta_1 + \frac{2}{3}\zeta_1$ as it should. We write $T \equiv T_\Phi$, $\dot{T} \equiv \partial_\eta T_\Phi$, and $T_q \equiv T_\Phi(q\eta/\sqrt{3})$ etc. in the kernels.

### 4.2 Second order: operator structure

At order $n \geq 2$ every component of $E^{(n)}_{\mu\nu} = n!\,[\epsilon^n]E_{\mu\nu}$ splits as

$$E^{(n)}_{\mu\nu} = \mathrm{Lin}_{\mu\nu}[X_n] + S^{(n)}_{\mu\nu}, \tag{42}$$

where Lin is the *same* linear operator as in (39) and $S^{(n)}_{\mu\nu}$ is built from lower-order fields ($S^{(2)}$ quadratic in $X_1$; $S^{(3)}$ contains $X_1X_2$ and $X_1^3$ terms; position-space expressions with 6–20 resp.

71–127 terms per component are in `position_sources.tex`). Decomposing $S^{(n)}_{\mu\nu}$ into scalar, vector and tensor parts (frame $\mathbf{k} = k\hat{\mathbf{z}}$, $\mathbf{q}$ in the $x$–$z$ plane),

$$S_{\rm TL} \equiv S_{zz} - \frac{1}{2}(S_{xx}+S_{yy}), \quad S_0^L \equiv S_{0z}, \quad S_{0i}^T \equiv (S_{0x}, S_{0y}), \quad S^V \equiv S_{xz}, \quad S^{TT} \equiv S_{xx} - S_{yy}, \quad S_{\rm tr} \equiv S_{ii}, \tag{43}$$

the ten equations become (all verified by the engine):

$$\begin{aligned}
\text{traceless scalar:} \quad & Q_n \equiv \Phi_n - \Psi_n = S_{\rm TL}/k^2 \quad \left( = -\frac{3}{2}\nabla^{-2}(\partial_i\partial_j\nabla^{-2} - \frac{1}{3}\delta_{ij})S_{ij} \text{ in position space}\right), \\
00: \quad & \delta_n = \frac{\eta^2}{3}S_{00} - \frac{2}{3}\eta^2 k^2 \Phi_n - 2\eta\Phi_n' - 2\Psi_n, \\
0i \text{ longitudinal:} \quad & v^L_{n,i} = \partial_i V_n, \qquad V_n = -\frac{\eta^2}{4}\Big[2(\Phi_n' + \Psi_n/\eta) + \frac{S_0^L}{ik}\Big], \\
0i \text{ transverse:} \quad & \frac{1}{2}\nabla^2\omega_{n,i} - \frac{4}{\eta^2}(\omega_{n,i} + v^T_{n,i}) = S^T_{0i} \quad [8\pi G a^2(\rho + P) = 4/\eta^2], \\
ij \text{ vector:} \quad & \mathcal{L}_\omega\, \omega_n = \frac{2}{ik}S^V, \qquad \mathcal{L}_\omega \equiv \partial_\eta + \frac{2}{\eta}, \\
ij \text{ TT:} \quad & \mathcal{L}_h\, h_{n,ij} = -2\Lambda_{ij,kl}(\hat{\mathbf{k}})S_{kl}, \qquad \mathcal{L}_h \equiv \partial_\eta^2 + \frac{2}{\eta}\partial_\eta + k^2, \\
\text{trace:} \quad & \mathcal{L}_\Phi\, \Phi_n = S_\Phi \equiv -\frac{S_{\rm tr} - S_{00}}{6} + \frac{Q_n'}{\eta} - \frac{k^2}{3}Q_n, \qquad \mathcal{L}_\Phi \equiv \partial_\eta^2 + \frac{4}{\eta}\partial_\eta + \frac{k^2}{3},
\end{aligned} \tag{44}$$

where $\Lambda_{ij,kl} = P_{ik}P_{jl} - \frac{1}{2}P_{ij}P_{kl}$, $P_{ij} = \delta_{ij} - \hat{k}_i\hat{k}_j$, and in the trace equation $\delta_n$ and $\Psi_n$ have been eliminated with the 00 and traceless equations. Thus at every order the scalar sector reduces to *one* forced oscillator $\mathcal{L}_\Phi\Phi_n = S_\Phi$, after which $\Psi_n, \delta_n, V_n$ are algebraic; the vector sector to a first-order equation; the tensor sector to a forced wave equation.

**Vector sector (relativistic Kelvin theorem).** Vorticity-freedom of $v^i$ is *not* implied at second order. What the equations imply (verified: the algebraic $\omega_2$ so obtained satisfies the $ij$-vector equation identically) is the circulation theorem for the isentropic radiation fluid, whose enthalpy per particle is $\propto T$:

$$\partial_{[i}[(1 + \delta_T)\,\hat{u}_{j]}] = 0. \tag{45}$$

Consequently $\omega_2$ is *algebraic* in the first-order fields and $v_2^T$ follows from the transverse $0i$ constraint; no vector Green's function is needed (the vector ODE has $G_\omega = (\eta'/\eta)^2$ and gives the same result).

**Kernels.** Second-order fields are written as

$$X_{2,\mathbf{k}}(\eta) = \int \frac{{\rm d}^3q}{(2\pi)^3}\, K_X(\mathbf{q}, \mathbf{p}; \eta)\, \zeta_{\mathbf{q}}\zeta_{\mathbf{p}}, \qquad \mathbf{p} \equiv \mathbf{k} - \mathbf{q}, \qquad \mu \equiv \hat{\mathbf{k}}\cdot\hat{\mathbf{q}} = \frac{k^2 + q^2 - p^2}{2kq}, \tag{46}$$

and all scalar kernels are given in the *invariant form* $K_X(k, q, p; \eta)$ (functions of the three magnitudes), equivalently in the $(k, q, \mu)$ form. With $T_q = T_\Phi(q\eta/\sqrt{3})$, $\dot{T}_q = \partial_\eta T_q$, the second-order

kernels obtained from (44) with $S^{(2)}$ are (full forms in App. A):

$$
\begin{aligned}
K_Q &= T_qT_p\Big[-4+\frac{6\mu q}{k}+\frac{3q^2}{k^2}-\frac{9\mu^2q^2}{k^2}\Big]+\Big[\eta(T_p\dot T_q+T_q\dot T_p)+\eta^2\dot T_q\dot T_p\Big]\Big[\frac{2\mu q}{k}+\frac{q^2}{k^2}-\frac{3\mu^2q^2}{k^2}\Big],\\
K_\omega &= \frac{2i}{k^2}\Big[\eta\,(q^2-p^2)\,T_qT_p+\eta^2(q^2T_q\dot T_p-p^2T_p\dot T_q)\Big]\quad(\omega_{2,i}=\int P_{ij}(\hat{\mathbf{k}})q_j\,K_\omega\,\zeta\zeta;\ \text{antisymmetric}\\
K_h &= 4\Big[3T_qT_p+\eta(\dot T_qT_p+T_q\dot T_p)+\eta^2\dot T_q\dot T_p\Big]=4\Big[2T_qT_p+(\eta\dot T+T)_q(\eta\dot T+T)_p\Big],\\
K_{S_{00}},\ K_{S_{0z}},\ K_{S_\Phi},\ K_{v^T}&:\ \text{App. A};\qquad v^T_{2,i}=\int P_{ij}q_jK_{v^T}\zeta\zeta,\qquad K_{v^T}=-\Big(\frac{\eta^2k^2}{8}+1\Big)K_\omega-\frac{\eta^2}{4}K_{S^T_{0i}}.
\end{aligned}
\tag{47}
$$

The tensor source $K_h$ is identical to the radiation-era kernel of Kohri–Terada [12] (their eq. (8)) and Baumann et al. [13] (their (11)–(12)) with $w=\frac{1}{3}$, $\Phi=\Psi$ (App. B, C1-9). $K_{S_\Phi}$ is finite as $\eta\to0$ (no $\eta^{-2}$ terms), so the particular integral of the trace equation vanishes at $\eta=0$; $\lim_{\eta\to0}\eta^2K_{S_{00}}=\frac{32}{3}$ and $K_Q(\eta\to0)$ is (31). The vector and tensor sources vanish as $\mathbf{p}\to-\mathbf{q}$ ($k\to0$) because of the projectors, which is what makes the $K\to0$ corner of the (22) loop integrable (Sec. 5.9).

**Green's functions and the regular solution.** With $x=k\eta/\sqrt{3}$, $x'=k\eta'/\sqrt{3}$,

$$
\begin{aligned}
u_1(x)&=\frac{\sin x-x\cos x}{x^3},\qquad u_2(x)=\frac{\cos x+x\sin x}{x^3},\qquad \hat g(x)\equiv3u_1(x)=-\frac{3}{2}T_\Phi(x)\ \to1,\qquad W\equiv u_1u_2'-u\\
G_\Phi(k;\eta,\eta')&=\frac{u_1(x')u_2(x)-u_1(x)u_2(x')}{W(\eta')}=\frac{\sqrt3}{k}\,\frac{x'}{x^3}\Big[(1+xx')\sin(x-x')-(x-x')\cos(x-x')\Big]\ \xrightarrow{k\to0}\ \frac{\eta'}{3}\Big(1-\frac{\eta'^3}{\eta^3}\\
G_h(k;\eta,\eta')&=\frac{\eta'}{\eta}\,\frac{\sin k(\eta-\eta')}{k},
\end{aligned}
\tag{48}
$$

satisfying $\mathcal{L}G=0$, $G(\eta,\eta)=0$, $\partial_\eta G|_{\eta'=\eta}=1$. The regular second-order scalar and tensor fields are then

$$
\begin{aligned}
\Phi_{2,\mathbf{k}}(\eta)&=\int\frac{\mathrm{d}^3q}{(2\pi)^3}\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}\,\mathcal{K}_{\Phi_2}(k,q,p;\eta),\qquad \mathcal{K}_{\Phi_2}\equiv\int_0^\eta\mathrm{d}\eta'\,G_\Phi(k;\eta,\eta')K_{S_\Phi}(k,q,p;\eta')+K^0_{\Phi_2}(k,q,p)\,\hat g(k\eta),\\
h_{2,ij,\mathbf{k}}(\eta)&=\Lambda_{ij,kl}(\hat{\mathbf{k}})\int\frac{\mathrm{d}^3q}{(2\pi)^3}q_kq_l\,\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}\,\mathcal{K}_h(k,q,p;\eta),\qquad \mathcal{K}_h\equiv\int_0^\eta\mathrm{d}\eta'\,G_h(k;\eta,\eta')K_h(k,q,p;\eta'),
\end{aligned}
\tag{49}
$$

with $K^0_{\Phi_2}$ the initial amplitude (32) fixed by Gaussianity, and $\Psi_2=\Phi_2-Q_2$, $\delta_2$, $V_2$ algebraic from (44). The lower limit 0 may be replaced by $\eta_i$ together with the homogeneous solution matched to the regular data at $\eta_i$ (Sec. 3.2). The time integrals $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$ are [unevaluated integral: $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$; closed forms in terms of Si, Ci of the Kohri–Terada type are being derived separately and will be quoted in Sec. 8].

**Checks at second order.** Bianchi propagation of the constraints (C1-5); exact dilation family (36) (C1-6); tensor source = Kohri–Terada (C1-9); superhorizon $\Phi_2$, $\Psi_2$ = Pitrou–Uzan–Bernardeau (C1-10); vorticity structure (45) (C1-7). An independent referee re-derivation (agent V3) is in progress and will be reported in Sec. 9.

### 4.3 Third order: contracted kernels

The (13) spectrum needs only the correlation of a third-order field with $\zeta_1$, i.e. the *q-contracted* third-order field,

$$\langle X_{3,\mathbf{k}}\,\zeta_{\mathbf{k}'}\rangle = (2\pi)^3\delta^3(\mathbf{k}+\mathbf{k}')\,P(k)\,X_{3c}(k,\eta), \qquad X_{3c}(k,\eta) = \int\frac{\mathrm{d}^3q}{(2\pi)^3}P(q)\,K_{3c,X}(k,q,\mu;\eta), \quad (50)$$

where the kernel sums the three Wick pairings (it is *not* symmetrised under $\mathbf{q}\to-\mathbf{q}$, so only $p=|\mathbf{k}-\mathbf{q}|$ appears). The contracted source $K_{3c,S}$ is built from $S^{(3)}$ by the rules

$$A_1B_1C_1 \;\to\; A_qB_{-q}C_k + A_qC_{-q}B_k + B_qC_{-q}A_k, \qquad A_1B_2 \;\to\; 2A_q\,K_B(-\mathbf{q},\mathbf{k}) + A_k\,\langle B_2\rangle(\eta), \quad (51)$$

in which $Q_2,\delta_2,V_2,v_2^T,\omega_2$ enter through their explicit kernels and $\Phi_2$, $h_2$ through the declared integrals $\mathcal{K}_{\Phi_2}(a,b,c;\eta)$, $\mathcal{K}_h(a,b,c;\eta)$ with $(a,b,c)=(|\mathbf{q}_1|,|\mathbf{q}_2|,|\mathbf{q}_1+\mathbf{q}_2|)$ kept as symbols. The mean-field terms

$$\langle X_2\rangle(\eta) = \int\frac{\mathrm{d}^3q}{(2\pi)^3}P(q)\,\mathrm{tad}_X(q,\eta), \qquad \mathrm{tad}_X(q,\eta)\equiv\langle K_X(\mathbf{q},-\mathbf{q};\eta)\rangle_{\mathrm{ang}}, \quad (52)$$

vanish for vectors, tensors and velocities by isotropy; for the scalars $Q_2, S_{00}, S_\Phi, \Phi_2, \delta_2$ the angle-averaged $\mathbf{k}\to0$ limits are given in closed form in App. A (e.g. $\mathrm{tad}_{\Phi_2}(q,\eta) = -\frac{8}{9} + \int_0^\eta \frac{\eta'}{3}(1-\frac{\eta'^3}{\eta^3})\mathrm{tad}_{S_\Phi}(q,\eta')\mathrm{d}\eta' = -\frac{8}{9} - \frac{28}{405}q^2\eta^2 + \ldots$), with the zero-mode caveat of Sec. 3.4.

The contracted third-order equations have the same operators (44):

$$Q_{3c} = K_{3c,Q}, \qquad \delta_{3c} = \frac{\eta^2}{3}K_{3c,S_{00}} - \frac{2}{3}\eta^2k^2\Phi_{3c} - 2\eta\Phi'_{3c} - 2\Psi_{3c}, \qquad V_{3c} = -\frac{\eta^2}{4}\Big[2(\Phi'_{3c}+\Psi_{3c}/\eta) + \frac{K_{3c,S_{0z}}}{ik}\Big],$$
$$\mathcal{L}_\Phi\Phi_{3c} = K_{3c,S_\Phi}, \qquad \Phi_{3c}(k,\eta) = \int\frac{\mathrm{d}^3q}{(2\pi)^3}P(q)\Big[\int_0^\eta \mathrm{d}\eta'\,G_\Phi(k;\eta,\eta')K_{3c,S_\Phi}(k,q,\mu;\eta') + C_3(k;q,\mu)\,\hat g(k\eta)\Big], \quad (53)$$

with $C_3$ the amplitude (35) fixed by $\zeta^{(3)}(0)=0$ and $K_{3c,S_\Phi}$ finite at $\eta\to0$. The contracted vector and tensor sources $(S_{0x}, S_{xz}, S_{xx}-S_{yy})$ are produced as well but integrate to zero over the $\mathbf{q}$ angles by isotropy ($S_{0y}, S_{yz}, S_{xy}$ vanish identically). The kernels $K_{3c,X}$ have 157–830 terms and are not reproduced; they are stored in machine-readable form (`third_order.pkl`, loader `kernels_api.load()['third']`, LaTeX dump `third_order.tex`). The $\eta'$ integral in (53) is [un-evaluated integral: $\int_0^\eta G_\Phi K_{3c,S_\Phi}$, which itself contains $\mathcal{K}_{\Phi_2}$ and $\mathcal{K}_h$ at the lower-order momenta]. The un-contracted kernel $K_3(\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3)$ is not needed for $C_\ell^{(13)}$ and has not been produced.

**Summary of what is settled.** First order: closed forms (41). Second order: all kernels in closed form except the two time integrals $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$, which are declared. Third order: contracted sources and amplitude $C_3$ in closed form; $\Phi_{3c}$ declared as one further nested time integral. The zero-mode convention for the tadpoles is a declared convention whose irrelevance for the observable is to be shown in Sec. 9.

# 5 Normalisation, loop structure and harmonic machinery

## 5.1 Normalisation bookkeeping (exact algebra)

Define $\Lambda \equiv -S$ and $M \equiv e^{\Lambda}$, expanded as in (5):

$$\Lambda = \epsilon L_1 + \frac{\epsilon^2}{2} L_2 + \frac{\epsilon^3}{6} L_3, \qquad M = 1 + \epsilon m_1 + \frac{\epsilon^2}{2} m_2 + \frac{\epsilon^3}{6} m_3, \qquad m_1 = L_1, \quad m_2 = L_2 + L_1^2, \quad m_3 = L_3 + 3L_1L_2 + L_1^3. \tag{54}$$

By Corollary 2.5, $1 + \Theta = M/\bar{M}$ with $\bar{M} \equiv \langle M \rangle_{\rm sky}$. Writing $\langle m \rangle_{\rm sky}$ for the sky monopole of a map ($\langle m \rangle_{\rm sky} = m_{00}/\sqrt{4\pi}$, a random variable correlated with the map) and expanding $1/\bar{M}$, one finds for $\ell \geq 1$

$$\boxed{\Theta^{(1)}_{\ell m} = m_{1,\ell m}, \qquad \Theta^{(2)}_{\ell m} = m_{2,\ell m} - 2\langle m_1 \rangle_{\rm sky} m_{1,\ell m}, \qquad \Theta^{(3)}_{\ell m} = m_{3,\ell m} - 3\langle m_1 \rangle_{\rm sky} m_{2,\ell m} - 3\langle m_2 \rangle_{\rm sky} m_{1,\ell m} + 6\langle m_1 \rangle^2_{\rm sk}} \tag{55}$$

(the monopole $\ell = 0$ of $\Theta$ vanishes identically). The measured dipole is removed afterwards, i.e. (10) is used for $\ell \geq 2$ only; by Sec. 2(iii) this is frame independent. All observer terms (the monopole $-[\delta_T + \Psi]_o$ and the dipole $\hat{\mathbf{n}} \cdot v_o$ of $L_1$, the clock shift $\delta\eta_o$, the observer position) are retained inside $L_1, L_2, L_3$ and are removed only through (55); products such as $\langle m_1 \rangle_{\rm sky} m_1$ and $L_1^2$ keep their imprint at $\ell \geq 2$.

## 5.2 Loop structure

Write each order as a direction-space kernel acting on the Gaussian field,

$$\Theta^{(r)}(\hat{\mathbf{n}}) = \int \prod_{i=1}^{r} \frac{\mathrm{d}^3 k_i}{(2\pi)^3}\, \zeta_{\mathbf{k}_1} \cdots \zeta_{\mathbf{k}_r}\, K_r(\mathbf{k}_1, \ldots, \mathbf{k}_r; \hat{\mathbf{n}}), \qquad K_{r,\ell m}(\mathbf{k}_1, \ldots) \equiv \int \mathrm{d}\Omega_{\hat{\mathbf{n}}} Y^*_{\ell m}(\hat{\mathbf{n}}) K_r, \tag{56}$$

with $K_r$ symmetric in its momenta (symmetrise if necessary). Wick's theorem for $\ell \geq 1$ then gives

$$\begin{aligned} C_\ell^{(11)} &= \frac{1}{2\ell+1} \sum_m \int \frac{\mathrm{d}^3 k}{(2\pi)^3} P(k)\, |K_{1,\ell m}(\mathbf{k})|^2, \\ C_\ell^{(22)} &= \frac{2}{2\ell+1} \sum_m \int \frac{\mathrm{d}^3 k_1\, \mathrm{d}^3 k_2}{(2\pi)^6} P(k_1) P(k_2)\, |K_{2,\ell m}(\mathbf{k}_1, \mathbf{k}_2)|^2, \\ C_\ell^{(13)} &= \frac{3}{2\ell+1} \sum_m \int \frac{\mathrm{d}^3 k}{(2\pi)^3} P(k)\, K_{1,\ell m}(\mathbf{k}) \Big[ \int \frac{\mathrm{d}^3 q}{(2\pi)^3} P(q)\, K_{3,\ell m}(\mathbf{k}, \mathbf{q}, -\mathbf{q}) \Big]^*. \end{aligned} \tag{57}$$

The factor 2 in $C^{(22)}$ counts the two connected pairings of a symmetric $K_2$; the third pairing $\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \rangle$ inside $\Theta^{(2)}$ has $\mathbf{k}_1 + \mathbf{k}_2 = 0$ and feeds only $\ell = 0$. The factor 3 in $C^{(13)}$ counts the three choices of which $\zeta$ inside $\Theta^{(3)}$ pairs with $\Theta^{(1)}$; the remaining pair is contracted at $\mathbf{q}_j = -\mathbf{q}_k$, so the "tadpole" configuration $K_3(\mathbf{k}, \mathbf{q}, -\mathbf{q})$ *is* the entire (13) term and must be kept (this is where the mean-field terms of Sec. 4.3 enter). Both prefactors have been re-derived independently and confirmed on the lensing control case below (App. B, C3-6). Equation (57) uses $\langle a^{(1)} a^{(3)*} \rangle$; only $\mathrm{Re}\, C_\ell^{(13)}$ enters (10). With the contracted fields of (50), $3 \int P(q) K_3(\mathbf{k}, \mathbf{q}, -\mathbf{q})$ is the $q$-contracted map $\Theta_c^{(3)}(\mathbf{k}; \hat{\mathbf{n}})$ built from $X_{3c}$ and the contraction rules (51).

### 5.3 Direction convention for the harmonic rules

A field on the background ray (8) (observer at $\bar{\mathbf{x}}_o = 0$) reads $f(\eta, \hat{\mathbf{n}}\chi) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} f_{\mathbf{k}}\, e^{+i\mathbf{k}\cdot\hat{\mathbf{n}}\chi} = \int \frac{\mathrm{d}^3k}{(2\pi)^3} f_{\mathbf{k}}\, e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi}$ with $\hat{\mathbf{e}} = -\hat{\mathbf{n}}$ the propagation direction. The rules below (and the code `cmb4/code/C3_harmonic/`) are written for maps expressed through $\hat{\mathbf{e}}$, i.e. with the exponent $e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi}$. Since $Y_{\ell m}(-\hat{\mathbf{n}}) = (-1)^\ell Y_{\ell m}(\hat{\mathbf{n}})$, the multipoles of the same map regarded as a function of $\hat{\mathbf{n}}$ differ by $(-1)^\ell$ at *every* order, so all $C_\ell^{(rs)}$ are unchanged; one convention must simply be used consistently for $\Theta^{(1)}, \Theta^{(2)}, \Theta^{(3)}$ together. In this subsection and the next, $\hat{\mathbf{n}}$ in the kernels of (56) is to be read as $\hat{\mathbf{e}}$. Rayleigh: $e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi} = 4\pi \sum_{\ell m} (-i)^\ell j_\ell(k\chi) Y_{\ell m}(\hat{\mathbf{e}}) Y^*_{\ell m}(\hat{\mathbf{k}})$. Gaunt integrals are

$$\mathcal{G}^{\ell\,\ell_1\ell_2}_{m\,m_1m_2} \equiv \int \mathrm{d}\Omega\, Y^*_{\ell m} Y_{\ell_1 m_1} Y_{\ell_2 m_2} = (-1)^m \sqrt{\frac{(2\ell+1)(2\ell_1+1)(2\ell_2+1)}{4\pi}} \begin{pmatrix} \ell & \ell_1 & \ell_2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell & \ell_1 & \ell_2 \\ -m & m_1 & m_2 \end{pmatrix}. \tag{58}$$

### 5.4 Type A: linear line-of-sight maps

A map of the form

$$\Theta_A(\hat{\mathbf{e}}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \zeta_{\mathbf{k}} \int_{\eta_*}^{\bar{\eta}_o} \mathrm{d}\eta \sum_{s\geq 0} g_s(k,\eta)\, (i\,\hat{\mathbf{e}}\cdot\hat{\mathbf{k}})^s\, e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi(\eta)} \tag{59}$$

(endpoint terms are $\delta(\eta-\eta_1)$ pieces of $g_s$; the monomials are written as $(i\hat{\mathbf{e}}\cdot\hat{\mathbf{k}})^s$ because $i\hat{\mathbf{k}} = \nabla/k$, so a real map has real $g_s$) has

$$\text{(A-}s\text{)} \qquad a_{\ell m} = 4\pi(-i)^\ell \int \frac{\mathrm{d}^3k}{(2\pi)^3} \zeta_{\mathbf{k}} Y^*_{\ell m}(\hat{\mathbf{k}})\, \Delta_\ell(k), \qquad \Delta_\ell(k) = \int \mathrm{d}\eta \sum_s g_s(k,\eta)\, (-1)^s j_\ell^{(s)}(k\chi(\eta)), \tag{60}$$

i.e. $(i\hat{\mathbf{e}}\cdot\hat{\mathbf{k}})^s e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi} \to (-\mathrm{d}/\mathrm{d}x)^s j_\ell(x)$, $x = k\chi$. Helicity pieces (vectors $\omega_i, v_i^T$ contracted with $\hat{\mathbf{e}}$, tensors $h_{ij}\hat{\mathbf{e}}^i\hat{\mathbf{e}}^j$): in the frame $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ with $e^{(\pm1)} = (\hat{\mathbf{x}} \pm i\hat{\mathbf{y}})/\sqrt{2}$, $e^{(\pm2)}_{ij} = e^{(\pm1)}_i e^{(\pm1)}_j$, $Q_{\pm1} = \hat{\mathbf{e}}\cdot e^{(\pm1)}$, $Q_{\pm2} = e^{(\pm2)}_{ij}\hat{\mathbf{e}}^i\hat{\mathbf{e}}^j$,

$$\text{(A-}\lambda\text{)} \qquad Q_\lambda e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi} = \sum_\ell (-i)^\ell \sqrt{4\pi(2\ell+1)}\, \hat{\Delta}^{(\lambda)}_\ell(x)\, Y_{\ell\lambda}(\hat{\mathbf{e}}), \quad \hat{\Delta}^{(\pm1)}_\ell = \mp i\sqrt{\frac{\ell(\ell+1)}{2}}\frac{j_\ell(x)}{x}, \quad \hat{\Delta}^{(\pm2)}_\ell = -\frac{1}{2}\sqrt{\frac{(\ell+2)}{(\ell-2)}} \tag{61}$$

and in a general frame $a^{(\lambda)}_{\ell m} = 4\pi(-i)^\ell \int \frac{\mathrm{d}^3k}{(2\pi)^3} \zeta^{(\lambda)}_{\mathbf{k}} (-1)^\lambda [{}_{-\lambda}Y_{\ell m}(\hat{\mathbf{k}})]^* \Delta^{(\lambda)}_\ell(k)$, $\Delta^{(\lambda)}_\ell = \int \mathrm{d}\eta\, g^{(\lambda)} \hat{\Delta}^{(\lambda)}_\ell(k\chi)$. Scalar, vector and tensor pieces of the same $\mathbf{k}$ do not cross-correlate. For two type-A maps

$$C_\ell^{AA'} = \frac{2}{\pi}\int k^2\mathrm{d}k\, P(k)\, \Delta^A_\ell(k)\Delta^{A'}_\ell(k)^*, \qquad \langle A\rangle_{\mathrm{sky}} = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\zeta_{\mathbf{k}}\Delta^A_0(k), \qquad \langle\langle A\rangle_{\mathrm{sky}}\langle A'\rangle_{\mathrm{sky}}\rangle = \frac{1}{2\pi^2}\int k^2\mathrm{d}k\, P\, \Delta^A_0\Delta^{A'}_0 \tag{62}$$

and $\langle\langle A\rangle_{\mathrm{sky}}\, a^{A'}_{\ell m}\rangle = 0$ for $\ell \geq 1$ by isotropy. The first-order map $m_1 = L_1$ of (24) is of type A (with $s \leq 1$ and endpoint terms), so $C_\ell^{(11)}$ is (62) with $A = A' = m_1$ (closed form: Sec. 8).

### 5.5 Type B: products of maps at the same point

For $B = AA'$ and $B' = CD$ with $A, A', C, D$ type-A maps, $b_{\ell m} = \sum \mathcal{G}^{\ell\ell_1\ell_2}_{mm_1m_2} w(\ell,\ell_1,\ell_2) a_{\ell_1 m_1} a'_{\ell_2 m_2}$, where $w = 1$ for a plain product and $w_\nabla = \frac{1}{2}[\ell_1(\ell_1+1) + \ell_2(\ell_2+1) - \ell(\ell+1)]$ for $\nabla_{\hat{\mathbf{n}}} A \cdot \nabla_{\hat{\mathbf{n}}} A'$ [16].

Wick's theorem with $\langle a^X_{\ell m} a^Y_{\ell' m'}\rangle = (-1)^m C^{XY}_\ell \delta_{\ell\ell'}\delta_{m,-m'}$ gives, for $\ell \geq 1$,

$$\text{(B1)} \qquad C^{BB'}_\ell = \sum_{\ell_1\ell_2} \frac{(2\ell_1+1)(2\ell_2+1)}{4\pi} \begin{pmatrix} \ell & \ell_1 & \ell_2 \\ 0 & 0 & 0 \end{pmatrix}^2 w\, w' \Big[ C^{AC}_{\ell_1} C^{A'D}_{\ell_2} + C^{AD}_{\ell_1} C^{A'C}_{\ell_2} \Big], \tag{63}$$

with $\ell + \ell_1 + \ell_2$ even; *both* Wick pairings appear and reduce to a single term with a factor 2 only when $A = A'$ (the third pairing $\langle AA'\rangle\langle CD\rangle$ feeds $\ell = 0$ only). Monopole×map terms, with $\langle X\rangle_{\rm sky}\cdot 1$ treated as an $\ell = 0$ map, give for $\ell \geq 1$

$$\begin{aligned}
&\text{(i)} \quad \frac{1}{2\ell+1}\sum_m \langle |\langle X\rangle_{\rm sky}\, y_{\ell m}|^2\rangle = \langle \langle X\rangle^2_{\rm sky}\rangle\, C^{YY}_\ell, \\
&\text{(ii)} \quad \frac{1}{2\ell+1}\sum_m \langle (AA')_{\ell m} (\langle X\rangle_{\rm sky}\, y_{\ell m})^*\rangle = \langle \langle A\rangle_{\rm sky}\langle X\rangle_{\rm sky}\rangle C^{A'Y}_\ell + \langle \langle A'\rangle_{\rm sky}\langle X\rangle_{\rm sky}\rangle C^{AY}_\ell, \\
&\text{(iii)} \quad \frac{1}{2\ell+1}\sum_m \langle (\langle X\rangle_{\rm sky} y_{\ell m})(\langle Z\rangle_{\rm sky} w_{\ell m})^*\rangle = \langle \langle X\rangle_{\rm sky}\langle Z\rangle_{\rm sky}\rangle C^{YW}_\ell.
\end{aligned} \tag{64}$$

Hence, for the type-B part of $m_2$ and with (55),

$$\frac{1}{4} C^{(22)}_\ell = \frac{1}{4}\Big[ C^{m_2 m_2}_\ell - 4\ \text{Re (ii)}|_{AA'\to m_2,\ X=Y=m_1} + 4\langle \langle m_1\rangle^2_{\rm sky}\rangle C^{(11)}_\ell \Big]. \tag{65}$$

*Type D* (nested, unequal-time products such as $\int \mathrm{d}\eta_1 \int_{\eta_1} \mathrm{d}\eta_2 W\, (\delta_{ij} - \hat e_i \hat e_j)\partial_i X(\eta_1)\partial_j Y(\eta_2)$, the lensing-type terms) reduce to (B1) applied to "shell maps" $\tilde X(\hat{\mathbf{e}}) = X(\eta_1, -\hat{\mathbf{e}}\chi_1)$ with unequal-time spectra $C^{xy}_\ell(\eta,\eta') = \frac{2}{\pi}\int k^2 \mathrm{d}k\, P\, x(k,\eta) y(k,\eta')^* j_\ell(k\chi) j_\ell(k\chi')$, using the exact identity $(\delta - \hat e\hat e)_{ij}\partial_i X \partial_j Y = (\chi_1\chi_2)^{-1}\nabla_{\hat{\mathbf{e}}}\tilde X \cdot \nabla_{\hat{\mathbf{e}}}\tilde Y$ (weight $w_\nabla$) and $\hat{\mathbf{e}}\cdot\partial X = -\partial_\chi \tilde X$ for radial contractions ($j_\ell \to -k j'_\ell$, weight 1).

### 5.6 Type C: convolution kernels

A genuinely second-order map

$$\Theta_C(\hat{\mathbf{e}}) = \int \frac{\mathrm{d}^3k_1 \mathrm{d}^3k_2}{(2\pi)^6} \zeta_{\mathbf{k}_1}\zeta_{\mathbf{k}_2} \int \mathrm{d}\eta\, K(k_1,k_2,K,\eta)\, \mathcal{Q}(\hat{\mathbf{e}};\hat{\mathbf{k}}_1,\hat{\mathbf{k}}_2,\hat{\mathbf{K}})\, e^{-i\mathbf{K}\cdot\hat{\mathbf{e}}\chi}, \qquad \mathbf{K} = \mathbf{k}_1 + \mathbf{k}_2, \tag{66}$$

with $\mathcal{Q} = \sum q_{abc}(k_1,k_2,\mu)(\hat{\mathbf{e}}\cdot\hat{\mathbf{k}}_1)^a(\hat{\mathbf{e}}\cdot\hat{\mathbf{k}}_2)^b(\hat{\mathbf{e}}\cdot\hat{\mathbf{K}})^c$ a polynomial of degree $d$ (transverse and TT projectors contracted with $\hat{\mathbf{e}}$ are of this form), has

$$K_{2,\ell m}(\mathbf{k}_1,\mathbf{k}_2) = \int \mathrm{d}\eta\, K \sum_{L=|\ell-d|}^{\ell+d} \sum_M 4\pi(-i)^L\, j_L(K\chi)\, Y^*_{LM}(\hat{\mathbf{K}})\, \langle \ell m|\mathcal{Q}|LM\rangle, \qquad \langle \ell m|\mathcal{Q}|LM\rangle = \int \mathrm{d}\Omega\, Y^*_{\ell m}\mathcal{Q}\, Y_{LM}, \tag{67}$$

a finite $3j$ sum generated by repeated use of $(\hat{\mathbf{e}}\cdot\hat v)Y_{\ell m} = \frac{4\pi}{3}\sum_\mu Y^*_{1\mu}(\hat v)\sum_{\ell'=\ell\pm1,m'} \mathcal{G}^{\ell'\ell 1}_{m' m\mu} Y_{\ell' m'}$. In the frame $\hat{\mathbf{K}} = \hat{\mathbf{z}}$, $\mathbf{k}_{1,2}$ in the $x$–$z$ plane ($\cos\alpha_1 = (k_1 + k_2\mu)/K$, $\cos\alpha_2 = (k_2 + k_1\mu)/K$, $\mu = \hat{\mathbf{k}}_1\cdot\hat{\mathbf{k}}_2$), $K_{2,\ell m} \to \tilde K_{\ell m}(k_1,k_2,\mu) = \int \mathrm{d}\eta\, K \sum_L (-i)^L \sqrt{4\pi(2L+1)} j_L(K\chi)\langle \ell m|\mathcal{Q}|L0\rangle$ with $|m| \leq d$, and isotropy reduces the angular integrals of (57) to

$$\begin{aligned}
&\text{(C1)} \quad C^{(22)}_\ell = \frac{1}{4\pi^4(2\ell+1)} \int k_1^2 \mathrm{d}k_1\, k_2^2 \mathrm{d}k_2\, P(k_1)P(k_2) \int_{-1}^1 \mathrm{d}\mu \sum_m |\tilde K_{\ell m}(k_1,k_2,\mu)|^2, \\
&\text{(C2)} \quad \frac{1}{2\ell+1}\sum_m \langle C_{\ell m} B^*_{\ell m}\rangle = \frac{1}{4\pi^4(2\ell+1)} \int k_1^2 \mathrm{d}k_1 k_2^2 \mathrm{d}k_2\, PP \int \mathrm{d}\mu \sum_m \tilde K^C_{\ell m}\tilde K^{B*}_{\ell m},
\end{aligned} \tag{68}$$

where for a type-B map $\tilde{K}^B_{\ell m} = \frac{1}{2}\sum \mathcal{G}^{\ell\ell_1\ell_2}_{mm_1m_2}[\tilde{A}_{\ell_1m_1}(k_1)\tilde{A}'_{\ell_2m_2}(k_2) + (A \leftrightarrow A')]$, $\tilde{A}_{\ell_1m_1}(k) = 4\pi(-i)^{\ell_1}Y^*_{\ell_1m_1}(\hat{\mathbf{k}})\Delta^A_{\ell_1}(k)$ at the frame angles. A non-symmetric kernel must be symmetrised as $\frac{1}{2}(K + K^{\mathrm{T}})$ *before* the $m$-sum. Two-time kernels use the double Rayleigh expansion instead of (67); (C1) is unchanged.

### 5.7 The (13) term

From (55),

$$\text{(E0)} \qquad C^{(13)}_\ell = T_{3,\ell} - 3T_{\langle 1\rangle_{\rm sky}2,\ell} - 3T_{\langle 2\rangle_{\rm sky}1,\ell} + 6\langle\langle m_1\rangle^2_{\rm sky}\rangle C^{(11)}_\ell, \qquad T_{3,\ell} \equiv \frac{1}{2\ell+1}\sum_m \langle m_{1,\ell m} m^*_{3,\ell m}\rangle, \tag{69}$$

and $T_{\langle 1\rangle_{\rm sky}2}$, $T_{\langle 2\rangle_{\rm sky}1}$ the corresponding cross terms with $\langle m_1\rangle_{\rm sky} m_2$, $\langle m_2\rangle_{\rm sky} m_1$. Every $q$-contracted map is a function of $\hat{\mathbf{e}}\cdot\hat{\mathbf{k}}$ only, hence a Legendre series handled by

$$\text{(A-}L\text{)} \qquad h(k,\eta)P_L(\hat{\mathbf{e}}\cdot\hat{\mathbf{k}})e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi} \;\Rightarrow\; \Delta_\ell = \int \mathrm{d}\eta\, h \sum_{\Lambda=|\ell-L|}^{\ell+L} (2\Lambda+1)\begin{pmatrix}\ell & L & \Lambda\\ 0 & 0 & 0\end{pmatrix}^2 (-i)^{\Lambda-\ell} j_\Lambda(k\chi), \tag{70}$$

and the $\mathbf{q}$-angular reduction

$$\text{(E1)} \qquad \int \frac{\mathrm{d}\Omega_q}{4\pi} g(\hat{\mathbf{q}}\cdot\hat{\mathbf{k}})(\hat{\mathbf{e}}\cdot\hat{\mathbf{q}})^b e^{-i\mathbf{q}\cdot\hat{\mathbf{e}}\Delta\chi} = \sum_{L'}(-i)^{L'}(2L'+1)j_{L'}(q\Delta\chi)\sum_{L''}\beta_{L'bL''}\, g_{L''}\, P_{L''}(\hat{\mathbf{e}}\cdot\hat{\mathbf{k}}), \tag{71}$$

with $g_{L''} = \frac{1}{2}\int \mathrm{d}\mu\, P_{L''}(\mu)g(\mu)$ and $x^b P_{L'} = \sum_{L''}\beta_{L'bL''}P_{L''}$; for $\Delta\chi = 0$ only $L' = 0$ survives. The three structures in $m_3$ are then:

(iii) genuine third-order kernels: $\Theta^{(3)}_c(\mathbf{k};\hat{\mathbf{e}}) = [\int \frac{q^2\mathrm{d}q}{2\pi^2}P(q)\sum_L h_L(k,q,\eta)P_L(\hat{\mathbf{e}}\cdot\hat{\mathbf{k}})]e^{-i\mathbf{k}\cdot\hat{\mathbf{e}}\chi}$ and $T^{(\mathrm{iii})}_{3,\ell} = \frac{2}{\pi}\int k^2\mathrm{d}k\, P\,\Delta^{m_1}_\ell(k)\Delta^{(3c)}_\ell(k)^*$ with $\Delta^{(3c)}_\ell$ from (A-$L$) (the factor 3 of (57) is already inside the contracted kernel);

(i) products of three type-A maps $ABC$: $T^{(\mathrm{i})}_{3,\ell} = C^{m_1A}_\ell \xi^{BC}(0) + C^{m_1B}_\ell\xi^{AC}(0) + C^{m_1C}_\ell\xi^{AB}(0)$, $\xi^{XY}(0) = \sum_{\ell'}\frac{2\ell'+1}{4\pi}C^{XY}_{\ell'}$ (with $\xi^{\nabla B\cdot\nabla C}(0) = \sum\frac{\ell'(\ell'+1)(2\ell'+1)}{4\pi}C^{BC}_{\ell'}$ and $\nabla B\nabla C : \nabla\nabla A \to -\ell(\ell+1)R_{BC}C^{m_1A}_\ell$);

(ii) type-A × type-C products $A\cdot C$: $T^{(\mathrm{ii})}_{3,\ell} = C^{m_1A}_\ell\langle\!\langle C\rangle\!\rangle + 2\cdot\frac{2}{\pi}\int k^2\mathrm{d}k\, P\,\Delta^{m_1}_\ell\Delta^{AC*}_\ell$, $\langle\!\langle C\rangle\!\rangle = \int\frac{\mathrm{d}^3k}{(2\pi)^3}P(k)K^C(\mathbf{k},-\mathbf{k};\hat{\mathbf{e}})$ and $\Delta^{AC}_\ell$ the transfer function of $\int\frac{\mathrm{d}^3q}{(2\pi)^3}P(q)K^A(\mathbf{q};\hat{\mathbf{e}})K^C(\mathbf{k},-\mathbf{q};\hat{\mathbf{e}})$ obtained from (E1) with $\Delta\chi = \chi_A - \chi_C$ and (A-$L$).

The normalisation pieces, with $M_2(\mathbf{q}_1,\mathbf{q}_2) \equiv K_{2,00}(\mathbf{q}_1,\mathbf{q}_2)/\sqrt{4\pi}$ the sky-average kernel of $m_2$, are

$$\text{(E2)} \quad T_{\langle 2\rangle_{\rm sky}1,\ell} = \langle\!\langle m_2\rangle\!\rangle C^{(11)}_\ell + \frac{1}{\pi^3}\int k^2\mathrm{d}k\, k'^2\mathrm{d}k'\, PP\,\Delta_\ell(k)\Delta_\ell(k')^*\int\mathrm{d}\mu\, P_\ell(\mu)M_2(k\hat{\mathbf{k}},-k'\hat{\mathbf{k}}')^*, \quad \langle\!\langle m_2\rangle\!\rangle = \frac{1}{2\pi^2}\int k^2$$

$$\text{(E3)} \quad T_{\langle 1\rangle_{\rm sky}2,\ell} = \frac{2}{2\pi^2}\int q^2\mathrm{d}q\, P(q)\Delta_0(q)\,\frac{2}{\pi}\int k^2\mathrm{d}k\, P(k)\Delta_\ell(k)\bar{\Delta}^{m_2}_\ell(k;q)^*, \qquad \bar{\Delta}_\ell(k;q) = \frac{\frac{1}{2}\int\mathrm{d}\mu\, K_{2,\ell0}(k\hat{\mathbf{z}},\mathbf{q})}{(-i)^\ell\sqrt{4\pi(2\ell+1)}}, \tag{72}$$

and the same objects give the cross term of (65). Formulas (E2), (E3) and $T^{(\mathrm{ii})}$ are derived by Wick counting consistent with the verified ones; (E3) collapses to (64)(ii) for a type-B $m_2$ (analytic check).

[**To be supplied:** the explicit $h_L$, $K^C$, $M_2$ of our map are supplied in Secs. 6–7 and assembled in Sec. 8.]

### 5.8 Control case: lensed $C_\ell$

For the remapping $\tilde{\Theta} = \Theta + \nabla\phi\cdot\nabla\Theta + \frac{1}{2}\nabla^i\phi\nabla^j\phi\nabla_i\nabla_j\Theta$ with $\Theta = \epsilon\Theta_1$, $\phi = \epsilon\phi_1$ one has $\Theta^{(2)} = 2\nabla\phi\cdot\nabla\Theta$, $\Theta^{(3)} = 3\nabla\phi\nabla\phi : \nabla\nabla\Theta$, and the bookkeeping (10) with (B1) and the (i)-type pairing reproduces exactly

$$\frac{1}{4}C_\ell^{(22)} = \sum_{\ell_1\ell_2}\frac{F^2_{\ell\ell_1\ell_2}}{2\ell+1}\Big[C^{\phi\phi}_{\ell_1}C^{\Theta\Theta}_{\ell_2} + C^{\phi\Theta}_{\ell_1}C^{\Theta\phi}_{\ell_2}\Big], \qquad \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)} = -\ell(\ell+1)R\,C_\ell^{\Theta\Theta}, \qquad R = \frac{1}{2}\langle|\nabla\phi|^2\rangle, \tag{73}$$

with $F$ the coupling of Hu [16]: these are Hu's (62)–(63) [16, 17] (the $C^{\phi\Theta}$ term is the extension to correlated $\phi,\Theta$, relevant here because the deflection potential correlates with SW/ISW). This was verified algebraically ($2\times10^{-16}$) and by Monte Carlo on the sphere (App. B, C3-5), validating the prefactors of (57), the two-pairing structure of (B1) and the monopole×map rules.

### 5.9 Monochromatic reduction: definitions

The monochromatic primordial spectrum is defined with the variance $\langle\zeta_1^2\rangle = \sigma^2$ held fixed as $k_0$ varies:

$$P(k) = \frac{2\pi^2\sigma^2}{k_0^2}\,\delta(k-k_0) \quad\Longleftrightarrow\quad \langle\zeta_1^2(\mathbf{x})\rangle = \sigma^2, \qquad \int k^2\mathrm{d}k\,P\,F(k) \to 2\pi^2\sigma^2F(k_0), \qquad \int\frac{\mathrm{d}^3k}{(2\pi)^3}P\,F(\mathbf{k}) \to \sigma^2\langle F(k_0\hat{\mathbf{k}}) \tag{74}$$

The expansion parameter is then $\epsilon^2\sigma^2$. Under (74) the general formulas collapse to (all transfer functions at $k_0$):

$$C_\ell^{AA'} = 4\pi\sigma^2\Delta_\ell^A\Delta_\ell^{A'*}, \qquad \langle\langle A\rangle_{\mathrm{sky}}\langle A'\rangle_{\mathrm{sky}}\rangle = \sigma^2\Delta_0^A\Delta_0^{A'}, \qquad \xi^{XY}(0) = \sigma^2\sum_{\ell'}(2\ell'+1)\Delta_{\ell'}^X\Delta_{\ell'}^{Y*};$$

$$\text{(B1)}:\ C_\ell^{BB'} = (4\pi\sigma^2)^2\sum_{\ell_1\ell_2}\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi}\begin{pmatrix}\ell & \ell_1 & \ell_2\\ 0 & 0 & 0\end{pmatrix}^2 ww'[\Delta^A_{\ell_1}\Delta^C_{\ell_1}\Delta^{A'}_{\ell_2}\Delta^D_{\ell_2} + \Delta^A_{\ell_1}\Delta^D_{\ell_1}\Delta^{A'}_{\ell_2}\Delta^C_{\ell_2}] \ \text{(effectively fini}$$

$$\text{(C1)}:\ C_\ell^{(22)} = \frac{\sigma^4}{2\ell+1}\int_{-1}^{1}\mathrm{d}\mu\sum_{|m|\le d}|\tilde{K}_{\ell m}(k_0,k_0,\mu)|^2, \quad K = k_0\sqrt{2(1+\mu)},\ \cos\alpha_1 = \cos\alpha_2 = \sqrt{(1+\mu)/2};$$

$$\text{(13)(iii)}:\ T_{3,\ell} = 12\pi\sigma^2\Delta_\ell\Delta_\ell^{(3c)*} \text{ with the } \frac{1}{3} \text{ of (10) still to be applied, } \Delta_\ell^{(3c)} = \sigma^2\sum_L h_L(k_0,k_0)\sum_\Lambda(2\Lambda+1)\begin{pmatrix}\ell & 1\\ 0 & 0\end{pmatrix}$$

$$h_L = \frac{1}{2}\int\mathrm{d}\mu\,P_L(\mu)[K_3(k_0,k_0,\mu)\times\mathcal{Q}_3\text{-coefficients}], \quad |\mathbf{k}\pm\mathbf{q}| = k_0\sqrt{2(1\pm\mu)};$$

$$\langle\langle m_1\rangle^2_{\mathrm{sky}}\rangle = \sigma^2\Delta_0^2, \quad \langle\!\langle m_2\rangle\!\rangle = \sigma^2M_2(k_0\hat{\mathbf{k}},-k_0\hat{\mathbf{k}}), \quad \text{(E2) cross} = 4\pi\sigma^4|\Delta_\ell|^2\int\mathrm{d}\mu\,P_\ell(\mu)M_2(k_0\hat{\mathbf{k}},-k_0\hat{\mathbf{k}}')^*, \quad \text{(E3)} = 8\pi \tag{75}$$

(The factor $12\pi\sigma^2 = 3\times4\pi\sigma^2$ in (13)(iii) assumes a symmetrised $K_3$ with the explicit pairing factor 3; with the contracted kernels of Sec. 4.3 the 3 is already included and the prefactor is $4\pi\sigma^2$.)

*Delicate points* to be addressed in Sec. 10: (1) $\mu \to -1$ in (C1), where $K \to 0$, $\mathrm{d}\mu = K\mathrm{d}K/k_0^2$ and $j_L(K\chi) \to \delta_{L0}$: a kernel $\sim K^{-p}$ converges only for $p < 1$, so every $1/K^2$ (Poisson inversion) must come with a vanishing numerator (the transverse/TT projections of $\mathbf{k}_1 \otimes \mathbf{k}_2$ do vanish at $\mathbf{k}_2 = -\mathbf{k}_1$; the scalar $K_Q$-type $1/K^2$ terms require an explicit check of the $K \to 0$ limit); (2) $\mu \to \pm 1$ in the (13) loop, where $|\mathbf{k} \mp \mathbf{q}| \to 0$ inside $K_3(\mathbf{k}, \mathbf{q}, -\mathbf{q})$ and $C_3$; (3) the $K = 0$ tadpoles $\langle\!\langle m_2 \rangle\!\rangle$, $\langle\!\langle C \rangle\!\rangle$ require kernels exactly at $\mathbf{K} = 0$ and enter only through (55), where the sky-mean normalisation cancels the direction-independent mean of $e^{\Lambda}$: individual tadpoles may be IR-singular while the combination is finite, so they are kept symbolic until combined (cf. Sec. 3.4); (4) $\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \rangle$ inside $\Theta^{(2)}$ feeds only $\ell = 0$, consistent with the monopole normalisation.

## 6 The second-order map $\Theta^{(2)}$

[**To be supplied:** Agent C2/assembly: explicit $L_1$, $L_2$ (hence $m_1$, $m_2$) as direction-space kernels $K_1(\mathbf{k}; \hat{\mathbf{n}})$, $K_2(\mathbf{k}_1, \mathbf{k}_2; \hat{\mathbf{n}})$ of the types A/B/C/D of Sec. 5, built from $S$ of Sec. 2 with the ray, observer-event and emission-point expansions; the observer clock shift $\delta\eta_o^{(1)}, \delta\eta_o^{(2)}$ and the observer position shift; the second-order fields of Sec. 4 inserted.]

## 7 The third-order map $\Theta^{(3)}$ in contracted form

[**To be supplied:** Agent C2/assembly: $L_3$ and $m_3 = L_3 + 3L_1L_2 + L_1^3$ in the $q$-contracted form $\Theta_{\mathrm{c}}^{(3)}(k; \hat{\mathbf{n}})$ of Sec. 5.7, including the mean-field (tadpole) pieces $\langle X_2 \rangle$ of Sec. 3.4, the third-order fields $X_{3c}$ of Sec. 4.3, and the third-order ray/observer/emission expansions.]

## 8 Assembly of $C_\ell$

[**To be supplied:** Assembly agent: $C_\ell^{(11)}$ (closed form, agent C5), $\frac{1}{4}C_\ell^{(22)}$ and $\frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}$ as fully specified integrals using the rules of Sec. 5; explicit statement of which pieces are evaluated in closed form and which remain as declared integrals (Sec. 11).]

## 9 Checks of the assembled observable

[**To be supplied:** Dilation check (constant $\zeta_1$ unobservable: tadpole independence, Sec. 3.4), gradient check (Mirbabayi–Zaldarriaga), comparison of the assembled $\Theta^{(1,2,3)}$ with the non-perturbative planar solver (agent N1, reference $a_{\ell 0}$ for a single plane wave), independent referee of the second/third-order kernels (agent V3), closed-form time integrals (agent C4). The checks already performed on the settled parts are listed in App. B.]

## 10 Monochromatic primordial spectrum

[**To be supplied:** Specialisation of the assembled $C_\ell$ to $P(k) = 2\pi^2\sigma^2\delta(k - k_0)/k_0^2$ (Sec. 5.9) with $\sigma^2 = \langle \zeta^2 \rangle$ held fixed; the single $\mu$-integral forms of Sec. 5.9; treatment of the $\mu \to -1$ ($K \to 0$) and $\mu \to \pm 1$ endpoints; exchange of limits justified analytically.]

# 11 Unresolved steps

[**To be supplied:** Complete list of declared-but-unevaluated integrals (time integrals over Green's functions, $\eta$-line-of-sight integrals, $k$-, $q$-, $\mu$-loop integrals), the tadpole zero-mode convention and its validation, any exchange of limit and integration not yet justified. Items flagged [unevaluated integral: ...] or [To be supplied] in the settled sections must be collected here.]

# A Explicit kernels (machine generated)

All expressions below are exported directly from the engine output of agent C1 (`cmb4/code/C1_einstein/second_o` `third_order_ic.pkl`; loader `kernels_api.load()`, `closed_form()` inserts $T_\Phi$). Notation: $T_q \equiv T_\Phi(q\eta/\sqrt{3})$, $\dot T_q \equiv \partial_\eta T_q$, $p = |\mathbf{k}-\mathbf{q}|$, $\mu = \hat{\mathbf{k}}\cdot\hat{\mathbf{q}} = (k^2+q^2-p^2)/(2kq)$; every second-order kernel is per $\zeta_{\mathbf{q}}\zeta_{\mathbf{p}}$ in the sense of (46). Sources are those of $E^{(2)}_{\mu\nu} = 2[\epsilon^2]E_{\mu\nu}$ in the frame $\mathbf{k} = k\hat{\mathbf{z}}$, $\mathbf{q}$ in the $x$–$z$ plane. The transfer functions (41) and the dictionary $\dot T_q = \partial_\eta T_\Phi$ with $\partial_\eta = (q/\sqrt{3})\,\mathrm{d}/\mathrm{d}x$ convert everything to closed form in $\sin, \cos$ of $q\eta/\sqrt{3}$, $p\eta/\sqrt{3}$.

## A.1 Second-order kernels

$K_Q$ **(traceless scalar,** $Q_2 = \Phi_2 - \Psi_2$**)**

$$
\begin{aligned}
&-\frac{T_pT_q\left(13k^4-6k^2p^2-6k^2q^2+9p^4-18p^2q^2+9q^4\right)}{4k^4}\\
&-\frac{T_p\dot T_q\eta\left(-k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{4k^4}\\
&-\frac{T_q\dot T_p\eta\left(-k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{4k^4}\\
&-\frac{\dot T_p\dot T_q\eta^2\left(-k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{4k^4}
\end{aligned}
\tag{76}
$$

$K_{S_{00}}$ **(source of the 00 constraint)**

$$
\begin{aligned}
&-\frac{2T_pT_q\left(\eta^2k^2+3\eta^2p^2+3\eta^2q^2-12\right)}{\eta^2}-T_p\dot T_q\eta\left(-k^2+p^2+q^2\right)\\
&-T_q\dot T_p\eta\left(-k^2+p^2+q^2\right)+\dot T_p\dot T_q\left(\eta^2k^2-\eta^2p^2-\eta^2q^2+6\right)
\end{aligned}
\tag{77}
$$

$K_{S_{0z}}$ **(longitudinal** $0i$ **source,** $S_0^L$**)**

$$
\begin{aligned}
&-\frac{2iT_pT_q\left(-\eta^2k^2p^2-\eta^2k^2q^2+\eta^2p^4-2\eta^2p^2q^2+\eta^2q^4-6k^2\right)}{3\eta k}\\
&-\frac{iT_p\dot T_q\left(-2\eta^2k^2p^2+2\eta^2p^4-2\eta^2p^2q^2-27k^2+9p^2-9q^2\right)}{3k}\\
&+\frac{iT_q\dot T_p\left(2\eta^2k^2q^2+2\eta^2p^2q^2-2\eta^2q^4+27k^2+9p^2-9q^2\right)}{3k}+4i\dot T_p\dot T_q\eta k
\end{aligned}
\tag{78}
$$

$K_{S_\Phi}$ **(source of $\mathcal{L}_\Phi \Phi_2 = S_\Phi$)**

$$\frac{T_p T_q \left(-7k^6 - 19k^4p^2 - 19k^4q^2 + 7k^2p^4 - 22k^2p^2q^2 + 7k^2q^4 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6\right)}{12k^4} \tag{79}$$
$$+ \frac{T_p \dot{T}_q \eta \left(k^2 + p^2\right)\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{12k^4}$$
$$+ \frac{T_q \dot{T}_p \eta \left(k^2 + q^2\right)\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{12k^4}$$
$$+ \frac{\dot{T}_p \dot{T}_q \left(-\eta^2k^6 - 2\eta^2k^4p^2 - 2\eta^2k^4q^2 + 3\eta^2k^2p^4 - 6\eta^2k^2p^2q^2 + 3\eta^2k^2q^4 + 12k^4 - 24k^2p^2 - 24k^2q^2 + 36p^4 - 72p^2q^2 -}{12k^4}$$

$K_\omega$ **(vector metric, $\omega_{2,i} = \int P_{ij} q_j K_\omega \zeta\zeta$)**

$$-\frac{2iT_pT_q\eta\,(p-q)\,(p+q)}{k^2} - \frac{2iT_p\dot{T}_q\eta^2p^2}{k^2} + \frac{2iT_q\dot{T}_p\eta^2q^2}{k^2} \tag{80}$$

$K_{v^T}$ **(transverse velocity, $v^T_{2,i} = \int P_{ij} q_j K_{v^T} \zeta\zeta$)**

$$-\frac{iT_pT_q\eta\,(p-q)\,(p+q)\left(\eta^2k^2 - 24\right)}{12k^2} - \frac{iT_p\dot{T}_q\eta^2\left(\eta^2k^2p^2 + 18k^2 - 24p^2\right)}{12k^2} \tag{81}$$
$$+ \frac{iT_q\dot{T}_p\eta^2\left(\eta^2k^2q^2 + 18k^2 - 24q^2\right)}{12k^2}$$

$K_h$ **(tensor source, $h_{2,ij} = \Lambda_{ij,kl} q_k q_l \int_0^\eta G_h K_h$)**

$$12T_pT_q + 4T_p\dot{T}_q\eta + 4T_q\dot{T}_p\eta + 4\dot{T}_p\dot{T}_q\eta^2 \tag{82}$$

$K^0_{\Phi_2}$ **(initial amplitude of $\Phi_2$, Eq. (32))**

$$-\frac{7k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{9k^4} \tag{83}$$

$K_Q(\eta \to 0)$

$$-\frac{13k^4 - 6k^2p^2 - 6k^2q^2 + 9p^4 - 18p^2q^2 + 9q^4}{9k^4} \tag{84}$$

**kernel of the local quadratic part $-\frac{1}{4}\delta_1^2 - 2\Phi_1^2$ of $\zeta^{(2)}$, Eq. (29) ($\to -\frac{4}{3}$ as $\eta \to 0$)**

$$-\frac{T_pT_q\left(\eta^4p^2q^2 + 3\eta^2p^2 + 3\eta^2q^2 + 27\right)}{9} - \frac{T_p\dot{T}_q\eta\left(\eta^2p^2 + 3\right)}{3} - \frac{T_q\dot{T}_p\eta\left(\eta^2q^2 + 3\right)}{3} - \dot{T}_p\dot{T}_q\eta^2 \tag{85}$$

**Algebraic second-order scalars** (with $S_{00}$, $S_{0z}$ the corresponding sources)

$$\delta_2 = \frac{S_{00}\eta^2}{3} - \frac{2\eta^2k^2\Phi_2(\eta)}{3} - 2\eta\frac{d}{d\eta}\Phi_2(\eta) - 2\Psi_2(\eta) \tag{86}$$

$$v_{2,z} = \frac{\eta\left(-S0z\eta - 2i\eta k\frac{d}{d\eta}\Phi_2(\eta) - 2ik\Psi_2(\eta)\right)}{4} \tag{87}$$

## A.2 Homogeneous solutions and Green's functions

$u_1$

$$\frac{3\sqrt{3}\left(-\frac{\sqrt{3}\eta k\cos\left(\frac{\sqrt{3}\eta k}{3}\right)}{3}+\sin\left(\frac{\sqrt{3}\eta k}{3}\right)\right)}{\eta^3k^3} \tag{88}$$

$u_2$

$$\frac{3\sqrt{3}\left(\frac{\sqrt{3}\eta k\sin\left(\frac{\sqrt{3}\eta k}{3}\right)}{3}+\cos\left(\frac{\sqrt{3}\eta k}{3}\right)\right)}{\eta^3k^3} \tag{89}$$

$\hat{g}=3u_1$

$$\frac{9\sqrt{3}\left(-\frac{\sqrt{3}\eta k\cos\left(\frac{\sqrt{3}\eta k}{3}\right)}{3}+\sin\left(\frac{\sqrt{3}\eta k}{3}\right)\right)}{\eta^3k^3} \tag{90}$$

$G_\Phi(k;\eta,\eta')$

$$\frac{\sqrt{3}\eta'\left(\eta'\eta k^2\sin\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right)+\sqrt{3}\eta' k\cos\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right)-\sqrt{3}\eta k\cos\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right)+3\sin\left(\frac{\sqrt{3}k(-\eta'+\eta)}{3}\right)\right)}{\eta^3k^3} \tag{91}$$

$G_h(k;\eta,\eta')$

$$\frac{\eta'\sin\left(k\left(-\eta'+\eta\right)\right)}{\eta k} \tag{92}$$

$G_\omega(\eta,\eta')$ **(for the ODE form of the vector equation)**

$$\frac{\eta'^2}{\eta^2} \tag{93}$$

## A.3 Third-order initial amplitude and intermediate limits

Per $P(q)\,\zeta_{\mathbf{k}}$ in the contracted sense of (50); $p^2=k^2-2kq\mu+q^2$.

$C_3(k;q,\mu)$**, Eq.** (35)

$$-\frac{16\left(2k^5+12k^4\mu q-42k^3\mu^2q^2-10k^3q^2+36k^2\mu^3q^3+32k^2\mu q^3+3k\mu^4q^4-42k\mu^2q^4+kq^4+8\mu q^5\right)}{9k\left(k^2-2k\mu q+q^2\right)^2} \tag{94}$$

$C_3$ **in invariant form** $(k,q,p)$

$$-\frac{35k^8+12k^6p^2-84k^6q^2+66k^4p^4-52k^4p^2q^2+66k^4q^4-84k^2p^6+84k^2p^4q^2+20k^2p^2q^4-20k^2q^6+3p^8-12p^6q^2}{9k^4p^4} \tag{95}$$

$Q_{3c}(\eta \to 0)$

$$-\frac{16q\left(44k^4\mu - 142k^3\mu^2q - 18k^3q + 108k^2\mu^3q^2 + 104k^2\mu q^2 + 9k\mu^4q^3 - 126k\mu^2q^3 - 3kq^3 + 24\mu q^4\right)}{9k\left(k^2 - 2k\mu q + q^2\right)^2} \tag{96}$$

$\lim_{\eta\to 0}\eta^2 K_{3c,S_{00}}$

$$-\frac{256kq\left(2k^2\mu - k\mu^2q - 3kq + 2\mu q^2\right)}{3\left(k^2 - 2k\mu q + q^2\right)^2} \tag{97}$$

### A.4 Tadpole (mean-field) functions

$\mathrm{tad}_X(q,\eta) = \langle K_X(\mathbf{q},-\mathbf{q};\eta)\rangle_{\mathrm{ang}}$, Eq. (52), angle-averaged $\mathbf{k}\to 0$ limits of the second-order kernels (convention of Sec. 3.4); small-$\eta$ behaviour in brackets.

$\mathrm{tad}_{Q_2}\ [-\frac{16}{9}+\frac{88}{405}q^2\eta^2+\ldots]$

$$\frac{8\left(\eta^6q^6\sin^2\left(\frac{\sqrt{3}\eta q}{3}\right)+\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+6\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-6\sqrt{3}\eta^3q^3\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+18\eta^2q^2\left(16\sin^2\left(\frac{\sqrt{3}\eta q}{3}\right)-\right.\right.}{15\eta^6q^6} \tag{98}$$

$\mathrm{tad}_{S_{00}}\ [\frac{32}{3\eta^2}-\frac{272}{45}q^2+\ldots]$

$$\frac{24\left(\frac{\eta^6q^6\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)}{2}-\frac{\eta^6q^6}{2}-2\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-21\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-6\eta^4q^4+36\sqrt{3}\eta^3q^3\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+126\eta^2q^2\right.}{\eta^8q^6} \tag{99}$$

$\mathrm{tad}_{S_\Phi}\ [-\frac{56}{81}q^2+\ldots]$

$$\frac{4\left(2\sqrt{3}\eta^7q^7\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)+12\eta^6q^6\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-30\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-684\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-270\eta^4q^4+1764\sqrt{3}\right.}{45\eta^8q^6} \tag{100}$$

$\mathrm{tad}_{\Phi_2}\ [-\frac{8}{9}-\frac{28}{405}q^2\eta^2+\ldots]$

$$\frac{-116\eta^6q^6+3\eta^3q^3\left(-2\sqrt{3}\eta^2q^2\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-18\eta q\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)+180\eta q+33\sqrt{3}\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)\right)+108\eta^2q^2\left(17\cos\left(\frac{2\sqrt{}\right.\right.}{45\eta^6q^6} \tag{101}$$

$\mathrm{tad}_{\delta_2}\ [\frac{16}{9}-\frac{472}{405}q^2\eta^2+\ldots]$

$$\frac{4\left(45\eta^6q^6\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)+19\eta^6q^6-186\sqrt{3}\eta^5q^5\sin\left(\frac{2\sqrt{3}\eta q}{3}\right)-1944\eta^4q^4\cos\left(\frac{2\sqrt{3}\eta q}{3}\right)-270\eta^4q^4+3879\sqrt{3}\eta^3q^3\sin\left(\frac{2\sqrt{}\right.\right.}{45\eta^6q^6} \tag{102}$$

### A.5 Not reproduced here

The contracted third-order source kernels $K_{3c,X}$ for $X \in \{Q, S_{00}, S_{0z}, S_\Phi, S_{0x}, S_{xz}, S_{xx} - S_{yy}\}$ (157–830 terms each, containing the declared integrals $\mathcal{K}_{\Phi_2}(a,b,c;\eta)$, $\mathcal{K}_h(a,b,c;\eta)$ as symbols) are in `third_order.pkl`/`third_order.tex`; the position-space sources $S^{(2)}_{\mu\nu}$, $S^{(3)}_{\mu\nu}$ in `position_sources.tex`; the full $E^{(n)}_{\mu\nu}$ in `E.tex`.

## B Verification ledger

Every check performed on the settled material, with its outcome. "Exact" means symbolic identity (sympy); numbers are maximal relative or absolute deviations. Code locations: V1 `code/V1_referee/`, C1 `code/C1_einstein/`, C2 `code/C2_observable/`, C3 `code/C3_harmonic/` (`run_all_tests.py`, log `test_log.txt`), N1 `code/N1_planar/`.

### Exact observable (referee V1; Sec. 2)

V1-1 Lemma 2.1 in a generic Lorentzian metric $g = G_0 + G_{1\alpha}x^\alpha$ (random rational coefficients, no symmetry), generic unit $u(x)$, generic null $p$: geodesic-derived $\mathrm{d}E/\mathrm{d}\lambda$ vs $-pp\nabla u$, decomposition, $\omega_{\mu\nu}p^\mu p^\nu = 0$, final identity: exact.

V1-2 Lemma 2.2: energy and Euler equations from $\nabla \cdot T = 0$ for general $\rho, P$; equivalence to (14) iff $\rho = 3P + c$, $P = CT^4 - c/4$; for $P = w\rho$, $\rho \propto T^4$ the identities hold iff $w = 1/3$: exact.

V1-3 Off-shell identity $\mathrm{d}\ln E/\mathrm{d}l - \mathrm{d}\ln T/\mathrm{d}l + \sigma_{ee} = -[(\theta + 3u\cdot\partial\ln T)/3 + (a + h\partial\ln T)\cdot e]$ at a generic point: exact (residual $-0.93$ off shell in a random instance, 0 on shell).

V1-4a FRW radiation: $\sigma = 0$, $T_* E_o/E_e/T_o = 1$ symbolically for all $p_i$.

V1-4b Bianchi I $\mathrm{d}s^2 = -\mathrm{d}t^2 + \sum t^{2q_i}\mathrm{d}x_i^2$ with a radiation test fluid $u = \partial_t$: $\mathrm{d}/\mathrm{d}t[\ln E - \ln T] + \sigma_{ee} = 0$ symbolically; integrated for Kasner $(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3})$ and non-Kasner $(\frac{1}{2}, \frac{1}{3}, \frac{1}{5})$, three directions, $t_e = 1 \to t_o = 5$: $|T^{\mathrm{exact}}_{\mathrm{obs}} - T_o e^{-S}| \lesssim 2 \times 10^{-31}$ with $O(1)$ direction dependence.

V1-4c Milne flow in Minkowski: $\theta = 3/\tau$, $\sigma = a = 0$, $T_{\mathrm{obs}} = T(o)$ isotropic: exact.

V1-5 Linear Poisson gauge: $\hat\theta, \hat a, \hat\sigma$ of (21); direct geodesic $\mathrm{d}\ln\hat E/\mathrm{d}\eta$ vs kinematic: agree; identity (23) off shell non-zero, on shell 0: exact. Sachs–Wolfe + Doppler + ISW with observer terms recovered.

V1-6 Tetrad rotation $R = \exp(\epsilon A)$ at $\ell = 2$ (exact spin-2 generators): $\sum_m |a_{\ell m}|^2$ invariant through $O(\epsilon^4)$ and $a^{(1)}$ invariant; $a^{(2)} \to a^{(2)} + 2Aa^{(1)}$; $C^{(22)}$ and $\mathrm{Re}\, C^{(13)}$ separately *not* invariant ($\sum|a_2|^2$: 321/25 vs $329107/1225 - 8798\sqrt{6}/1225$) while $\frac{1}{4}C^{(22)} + \frac{1}{3}\mathrm{Re}\, C^{(13)}$ is (107/60 both). Dipole power invariant.

### Exact observable, independent route (agent C2, milestone 1; Sec. 2.5)

C2-1 Kinematic identity $p^a p^b \nabla_a u_b = \sigma_{pp} + \theta(p\cdot p + E^2)/3 + Ea\cdot p$ for a generic (non-null) $p$, no equations of motion: zero at orders 1, 2, 3.

C2-2 Radiation identities (14) from $\hat\nabla_\mu \hat T^{\mu\nu} = 0$ alone (solved order by order for $\partial_\eta\delta_{T,n}$, $\partial_\eta v_n$): zero at orders 1, 2, 3.

C2-3 Integrand-level comparison of route (i) $[\ln(\hat{E}_o/\hat{E}_e) = -\int \Gamma^0_{kk} \mathrm{d}\eta + [\ln(-u\cdot k)]^o_e]$ with route (ii) $[\ln(\hat{T}_o/\hat{T}_e) - S]$ along an arbitrary perturbed ray $k^i = -n^i + \epsilon k^i_1 + \ldots$ with the non-affine geodesic equation substituted: residual $R \equiv 0$ at orders 1, 2 and 3 modulo conservation (14/70/340 terms before reduction).

C2-4 $-I_1 = -D[\delta_{T,1} + \Psi_1 - \hat{\mathbf{n}}\cdot v_1] + \Phi_1' + \Psi_1'$ with $I_1 = \sigma_{1,kk}$: exact (SPEC check (a)).

C2-5 Canonical triad (Gram–Schmidt w.r.t. $\hat{g}$, orthogonal to $\hat{u}$): $\hat{g}(e_a, e_b) = \delta_{ab}$, $\hat{g}(e_a, \hat{u}) = 0$ through order 3.

C2-6 Direction map (19): $k^\mu(\eta_o)$ null through order 3 (with $|\hat{\mathbf{n}}| = 1$); first-order geodesic right-hand side $G^i_1 = -\partial_i(\Phi + \Psi) + 2n^i\,\hat{\mathbf{n}}\cdot\nabla(\Phi + \Psi) - n^i(\Phi' + \Psi')$ consistent with the null condition $\hat{\mathbf{n}}\cdot k_1 = -(\Phi + \Psi)$.

### Einstein–fluid equations (agent C1; Secs. 3–4)

C1-1 Engine: $g\,g^{-1} = 1$ and $g_{\mu\nu}u^\mu u^\nu = -1$ to $O(\epsilon^3)$; background part of every $E_{\mu\nu}$ and of $\nabla_\mu T^{\mu\nu}$ vanishes identically.

C1-2 First-order equations reproduce the standard linearised Poisson-gauge equations (39); $\Psi_1 = \Phi_1$; $v_1^T = 0$; $\mathrm{curl}\, v_1 = 0$ identically.

C1-3 $\Phi_1'' + 4\Phi_1'/\eta + k^2\Phi_1/3 = 0$ verified; transfer functions (41) verified by substitution; conservation equations ($\nu = 0..3$) vanish identically on the solution; agreement with the lead's independent linear check ($T_\Phi$, $\delta_T \to \zeta/3$, $V \to \eta\zeta/3$).

C1-4 Second-order SVT structure (44) verified component by component ($S_{0y} = S_{yz} = S_{xy} = 0$ by reflection symmetry; vector/tensor kernels free of $\sqrt{1-\mu^2}$ after projection).

C1-5 Contracted Bianchi identity $g^{\mu\alpha}\nabla_\alpha E_{\mu\nu} + g_{\nu\beta}\nabla_\mu T^{\mu\beta} \equiv 0$ to $O(\epsilon^3)$ (hence the constraints propagate at every order).

C1-6 Exact dilation family (36) solves $E_{\mu\nu} = 0$ to $O(\epsilon^3)$ for arbitrary $A, B$; $k = 0$ second-order equations fix only $\delta_2 = -2\Psi_2 + \dfrac{32}{9}c^2$; angle-averaged $k \to 0$ limit of the regular solution lies on the family.

C1-7 Vector sector: $\mathrm{curl}\,\hat{u}_i \neq 0$ at second order; relativistic Kelvin (45) holds and the algebraic $\omega_2$ satisfies the $ij$-vector ODE identically.

C1-8 $\zeta^{(2)}$, $\zeta^{(3)}$ expansions (29)–(30); $\lim \eta^2 K_{S_{00}} = 32/3$; $K_Q(0)$ and $K^0_{\Phi_2}$ (31)–(32).

C1-9 Tensor source $K_h$ identical to Kohri–Terada (8) / Baumann et al. (11)–(12) with $w = 1/3$, $\Phi = \Psi$: numerically exact.

C1-10 Superhorizon $\Phi_2, \Psi_2$ (33)–(34) = Pitrou–Uzan–Bernardeau (3.12b,c) with $\mathcal{R}^{(2)} = -2\mathcal{R}^{(1)2}$: exact.

C1-11 Bartolo–Matarrese–Riotto II (7.30): local and gradient terms agree; velocity-quadrupole term differs ($27F$ vs our $3F$) because BMR use the late-time velocity; ours = PUB.

C1-12 Green's functions (48): $\mathcal{L}G = 0$, $G(\eta,\eta) = 0$, $\partial_\eta G|_{\eta'=\eta} = 1$ (re-checked for the rewritten form of $G_\Phi$ in this write-up).

C1-13 $K_{S_\Phi}$ and $K_{3c,S_\Phi}$ finite as $\eta \to 0$ (small-$\eta$ expansions), so particular integrals vanish at $\eta = 0$; contracted vector/tensor sources vanish after angular integration.

C1-14 Squeezed limit $C_3(q \to 0) = -32/9$ equals the prediction of the exact dilation family: pass.

### Harmonic machinery (agent C3; Sec. 5)

C3-1 Rule (A-$s$), $s = 0..3$; rule (A-$\lambda$), $\lambda = \pm1, \pm2$, in the $k$-frame and in a random frame (via $D$-matrices and spin-weighted harmonics): $10^{-14}$.

C3-2 $C^{(11)}$ normalisation $2/\pi$ via the real-space route (monochromatic $\xi = \sigma^2 |f|^2 j_0(k_0\chi|\hat{\mathbf{n}}_1 - \hat{\mathbf{n}}_2|) \Rightarrow C_\ell = 4\pi\sigma^2|f|^2 j_\ell^2$): $10^{-14}$.

C3-3 (B1), both pairings, plain and gradient weights, correlated maps, vs exact real-space Wick + Legendre transform: $2.6 \times 10^{-12}$; Monte Carlo (400 Gaussian realisations, exact band-limited SHT): plain product 6.6%, gradient product 4.1%, $\langle X \rangle_{\rm sky}$·map 4.5%, cross term (ii) 3.1% (statistical, excludes any factor-2 error).

C3-4 Type-D Fourier $\leftrightarrow$ angular-gradient identity (two radii): $2.9 \times 10^{-12}$.

C3-5 Lensing control: $\frac{1}{4}C^{(22)} = \sum F^2/(2\ell+1)[C^\phi C^\Theta + C^{\phi\Theta}C^{\Theta\phi}]$ algebraically ($2.5 \times 10^{-16}$); Monte Carlo of the Taylor-expanded remapping with correlated $\phi, \Theta$: (22) 2.3%, (13) $= -\ell(\ell+1)RC_\ell$ 4.6%; unlensed $C_\ell$ recovered 2.1%.

C3-6 SPEC prefactors re-derived: $C^{(22)}$ factor 2 and $C^{(13)}$ factor 3 confirmed; SPEC's $K_1^* K_3$ is the conjugate of $\langle a^{(1)} a^{(3)*} \rangle$ (irrelevant for Re).

C3-7 Type C: $K_{2,\ell m}$ finite $3j$ sums vs direct 2D quadrature (random $k_1, k_2$, two time nodes, degree 3): $1.1 \times 10^{-14}$; isotropy reduction random frame vs $\hat{\mathbf{K}} = \hat{z}$: $8.9 \times 10^{-15}$; (C1) $\equiv$ (B1) for product kernels (monochromatic): $3.7 \times 10^{-15}$; (C2) vs (B1): $3.6 \times 10^{-15}$.

C3-8 Rule (A-$L$): $3.2 \times 10^{-14}$; reduction (E1) with $b \le 3$, $a \le 2$, $\Delta\chi = 0$ and $\neq 0$: $\le 2.8 \times 10^{-15}$.

C3-9 Monochromatic normalisation $\langle \zeta^2 \rangle = \sigma^2$: $5.9 \times 10^{-7}$ (Gaussian approximation of the $\delta$-function).

C3-10 Not verified numerically (formulas only, Wick counting consistent with the verified ones): (E2), (E3), $T_3^{(\rm ii)}$; the collapse of (E3) to (64)(ii) for a type-B $m_2$ passes analytically.

### Non-perturbative arbiter (agent N1; planar symmetry)

N1-1 Exact planar GR + radiation solver (comoving-orthogonal gauge, lapse fixed algebraically by (14): $\alpha = e^{-\zeta}(AB^2)^{1/3}$, $T_{\rm phys} = 1/\alpha$), constraint violations $\sim 10^{-13}$: Theorem 2.3 verified non-perturbatively, $T_{\rm obs}/T_o = e^{-S}$ to $7 \times 10^{-15}$ at $\epsilon = 0.1$ ($\Theta_{\rm rms} = 0.08$).

N1-2 Linear limit of the planar solver vs the Poisson-gauge transfer functions (41) through a derived gauge transformation: pass. Reference $a_{\ell 0}[\Theta^{(1,2,3)}]$ for a single plane wave are stored for the assembly check (Sec. 9).

### Not yet checked / pending

Independent referee of the second- and third-order kernels (V3); dilation (constant-$\zeta$) and gradient-mode invariance of the assembled map (SPEC 9(d)); closed forms of $\mathcal{K}_{\Phi_2}$, $\mathcal{K}_h$ (C4); first-order $C_\ell^{(11)}$ closed form (C5); comparison of the assembled kernels with N1.

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